Artificial Intelligence Driven Meta-Analysis

Introduction to Meta Analysis



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Reminder for Group Project

1. Date of Final Report: Dec. 5, 2025
   - Deadline of Submission: 10.00 am
2. Date of Presentation: Dec. 12, 2025
   - Time : 10:00-13:00 and 19:00-22:00
   - Deadline of Submission: 10.00 am



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Major Requirements for Group Project

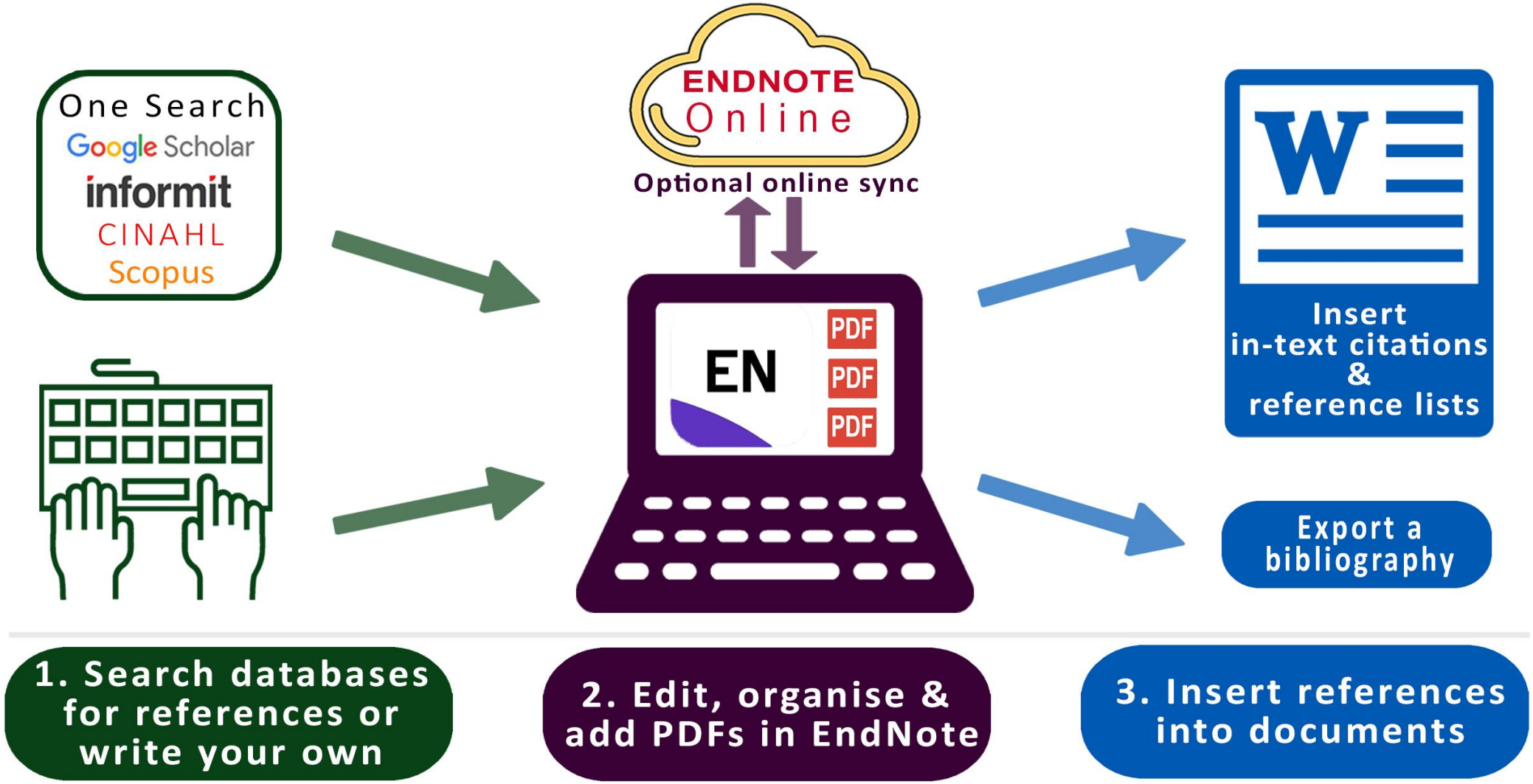
- Teams consist of **3 or 4 randomly** assigned members
- The project data will be released on **Sep 12, 2026** (today)
- Your team need to select **a topic** and utilize it to develop a detailed project.
- Teams are required to submit ***the final project reports with a Turnitin report, the ppt or pdf of the presentation slides, the work contribution and Gen AI documents***.
- Teams should select their **own presenter**.
- The project leader will submit **all project files**, and every member must submit a **peer evaluation for the entire team.**
- **For more details, please refer to the files in Canvas.**



Systematic Review And Meta-Analysis

A Step by Step guide Approach


Garbage in
Garbage OUT


One Search
Google Scholar
informit
CINAHL
Scopus
ENDNOTE
Online
Optional online sync
EN
PDF
PDF
PDF
Insert
in-text citations
&
reference lists
Export a
bibliography
1. Search databases for references or write your own
2. Edit, organise & add PDFs in EndNote
3. Insert references into documents

Contents

Systematic Review context

Managing downloads

De-duplication

Screening

Find full text

Backups

Sharing whole libraries or groups with collaborators

EndNote helps with…

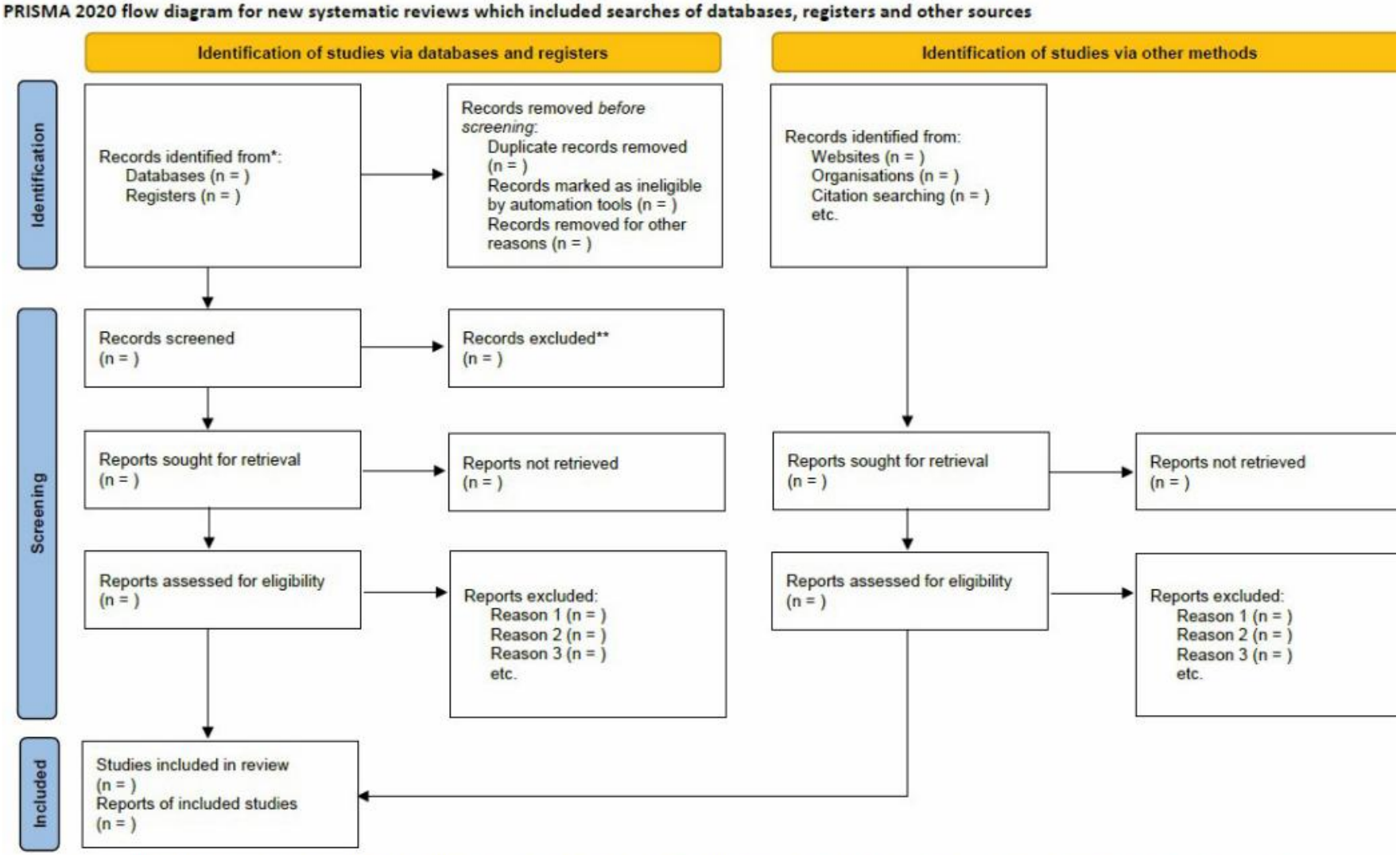
Managing downloads

(De-duplication)

(Screening)

Find full text

Citations in write-up



*Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/registers).
**If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools.

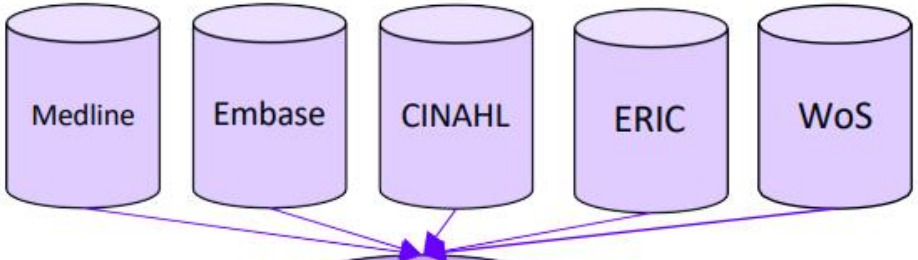
From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372: doi: 10.1136/bmj.n71. For more information, visit: http://www.prisma-statement.org/

Workflow

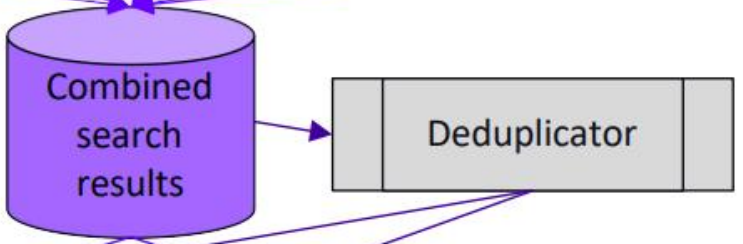
(Each container represents a separate EndNote library)

Create and keep an EndNote Library for each batch of search results

This workflow ensures you have a backup and record for search results at each stage

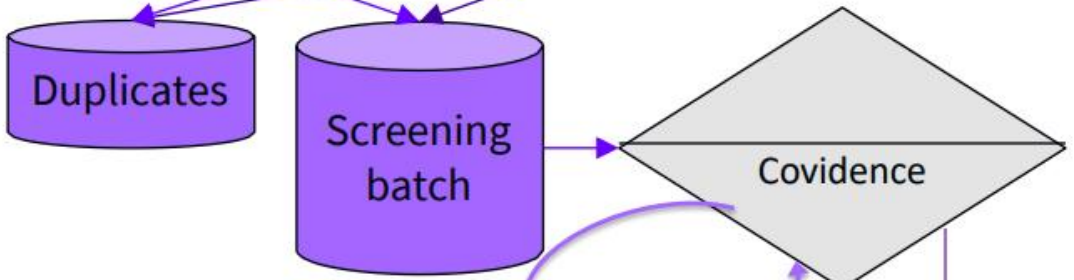


Copy and paste results into a combined EndNote library for all databases



Run a de-duplication, either within EndNote or an external tool like the SR Accelerator's Deduplicator tool

Don't delete duplicates – keep them set aside either in a group or separate EndNote Library



Screen results either in EndNote or using an external tool like Covidence

EndNote can quicken getting the full text journal articles



Use EndNote for inserting references into final write-up

Managing downloads

- Bulk downloads and keeping track of results numbers
- A cautious approach that provides backups
  - separate libraries for each results set
  - combined library of all results (back-up)
  - working copy of combined library for deduplication and screening

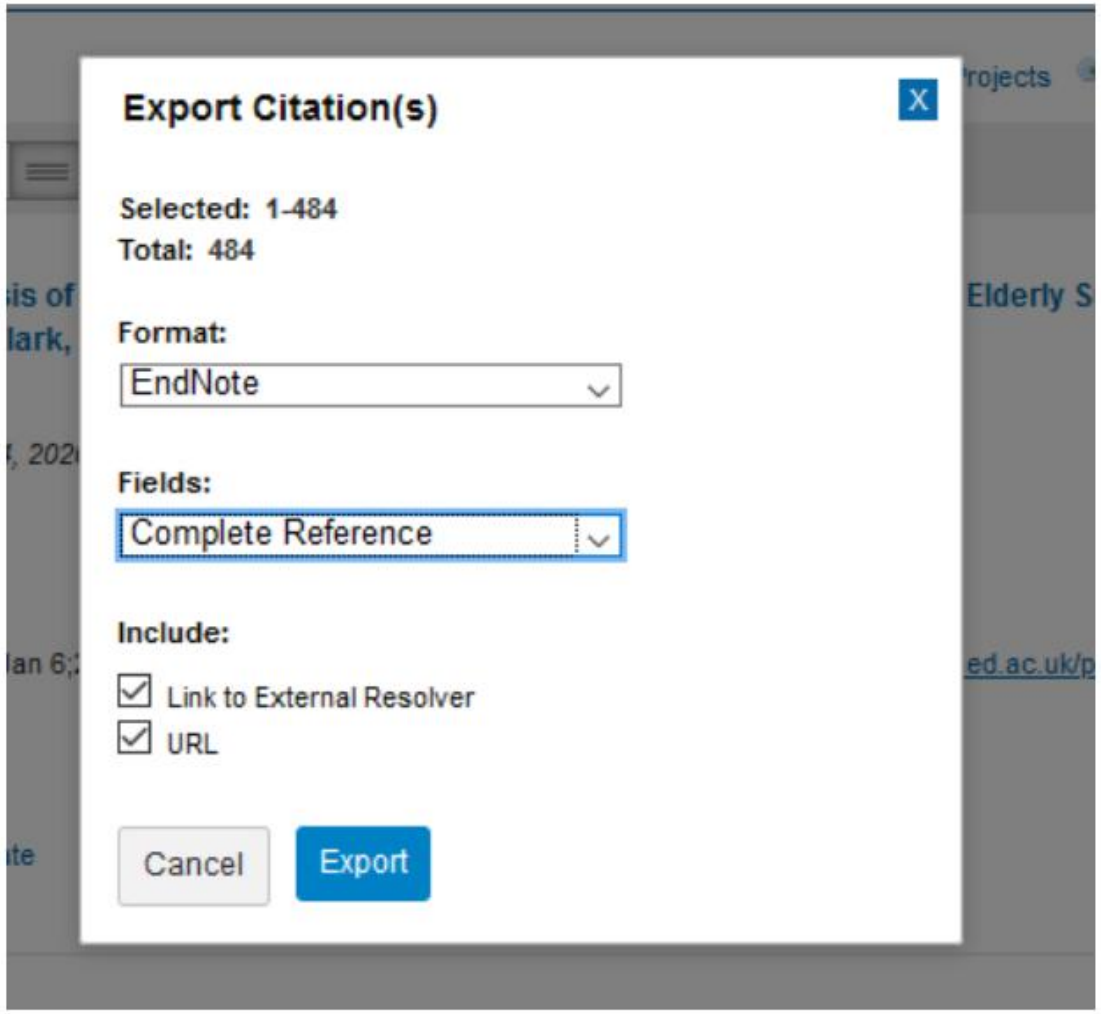
Bulk downloads

Screening against criteria for inclusion is best done offline

Use bulk downloads into EndNote

Be sure to note results numbers and check against import numbers

Be sure to select download options that include e.g. abstracts and DOIs, to make screening phases easier



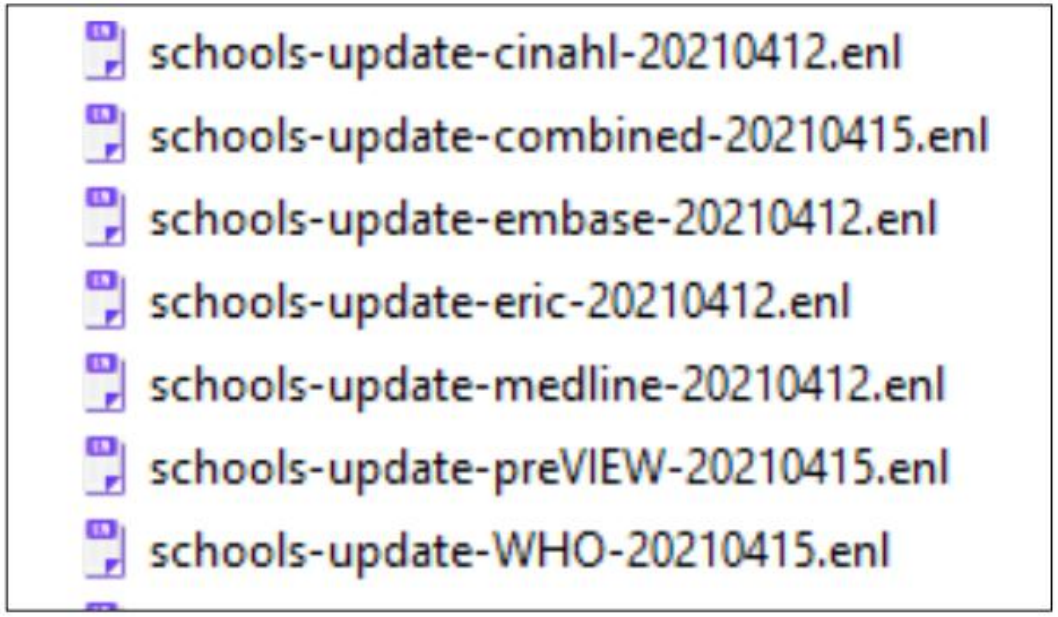
Libraries for each results set

This project example had six database results sets

One combined results library, from which duplicate results were copied in to separate library

Allows each set of results to be checked later if needed

Allows easy rescue if combined library is lost/corrupted



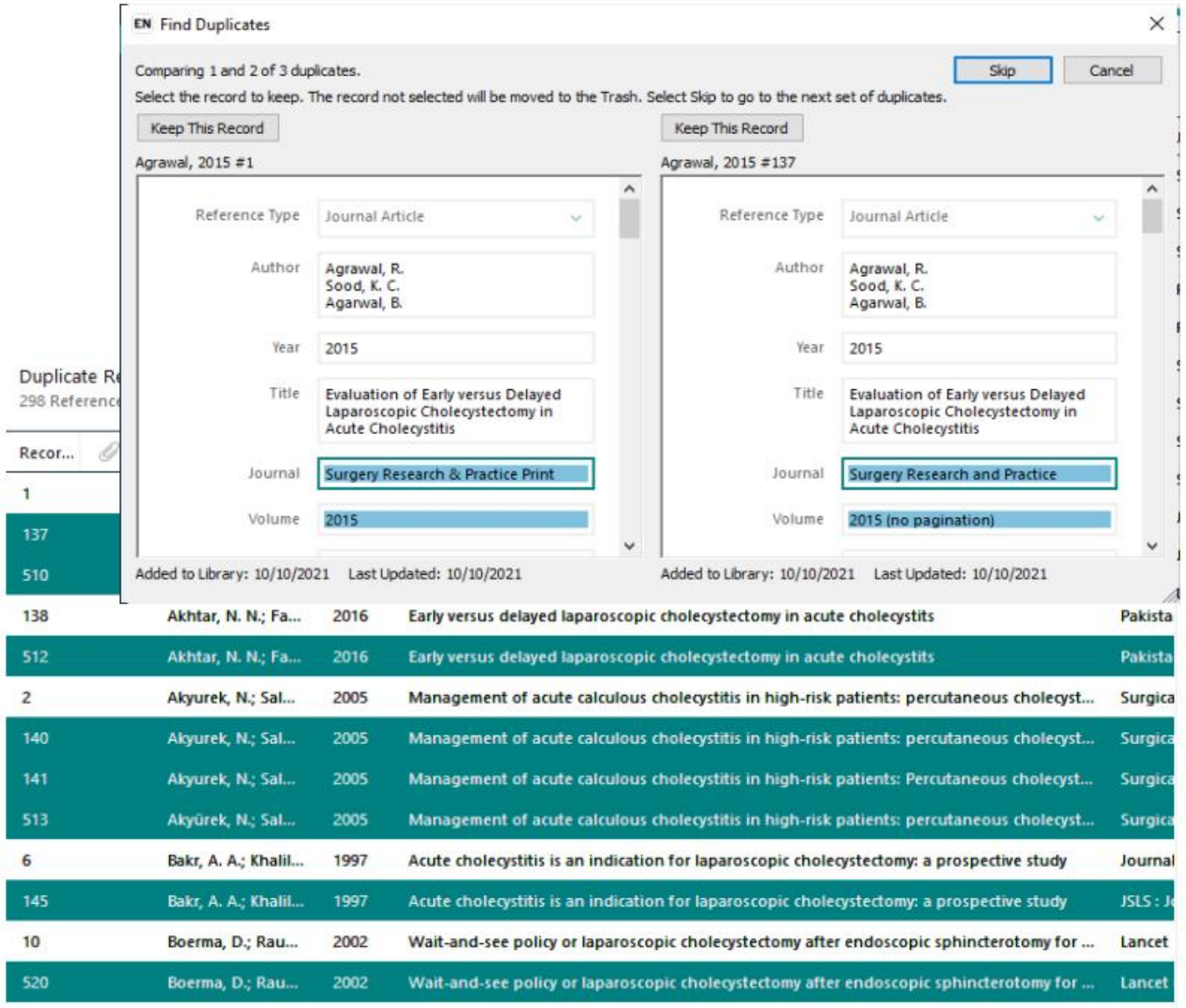
De-duplication 1

EndNote can be used for deduplication prior to screening

If you have a lot of duplicates, close the comparison window and screen list, to save time

Beware! The duplicates list shows both first copy (not highlighted) and duplicate copies (highlighted), so take care only to move duplicates.

Note: EndNote will miss duplicates with the default criteria. Consider adjusting criteria, but check carefully items identified as duplicate



De-duplication 2

External tools:

Deduplicator available from the Systematic Review Accelerator

https://sr-accelerator.com/#/deduplicator



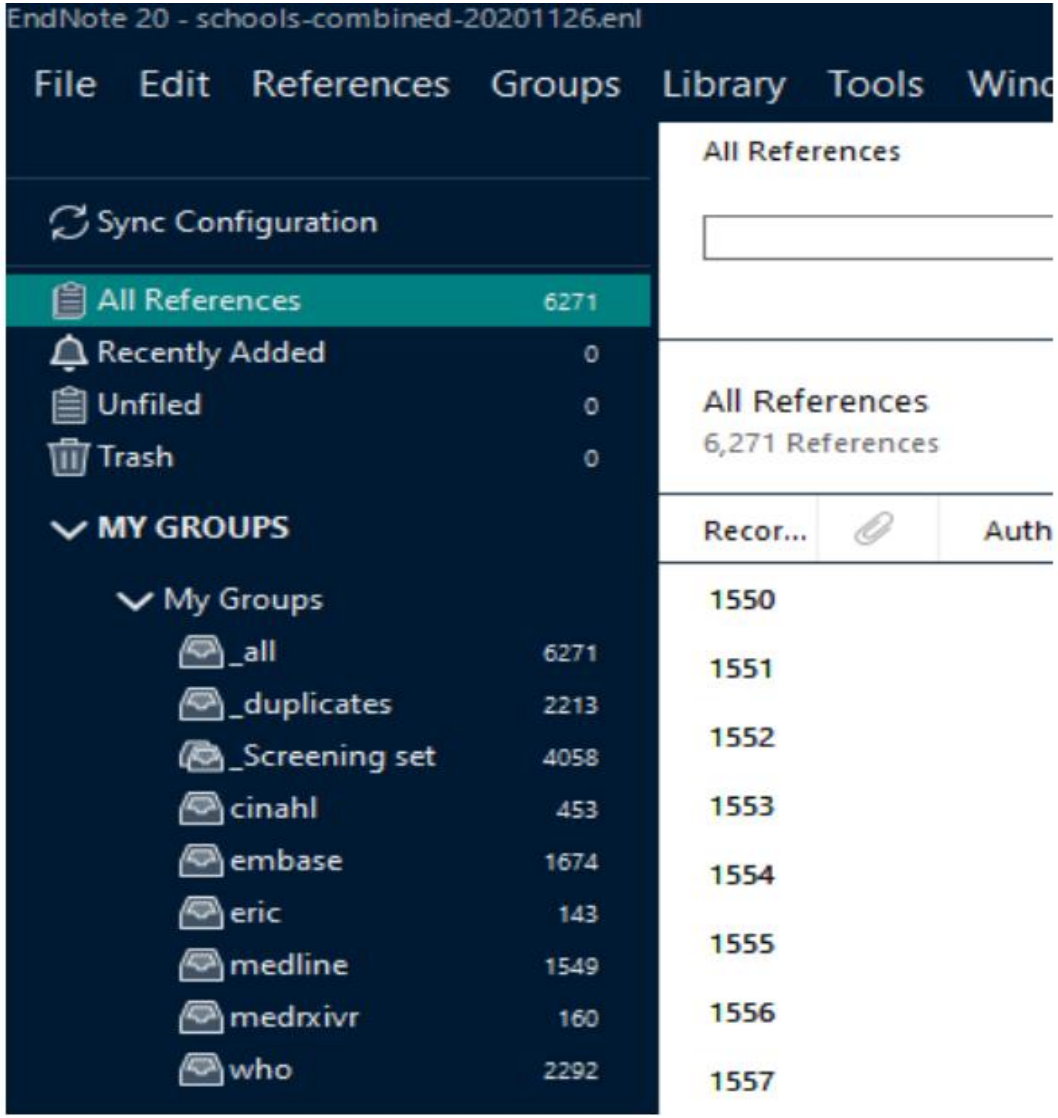
Covidence, which you may want to use for screening, also checks for duplicates as part of upload process.



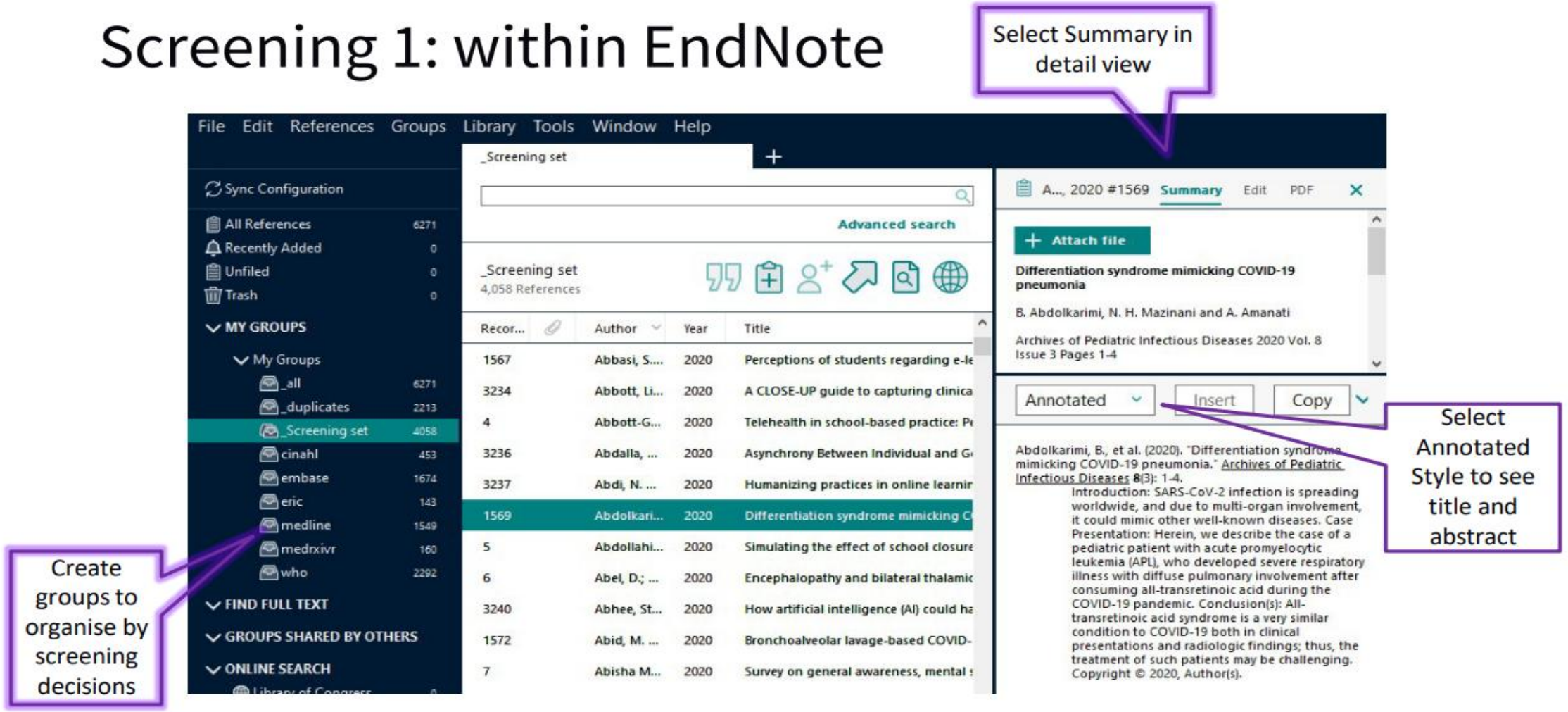
Using Groups

EndNote groups can be used to organise the records

- Keep track of screening decisions
- Keep track of database/origin
- Plan workload in team

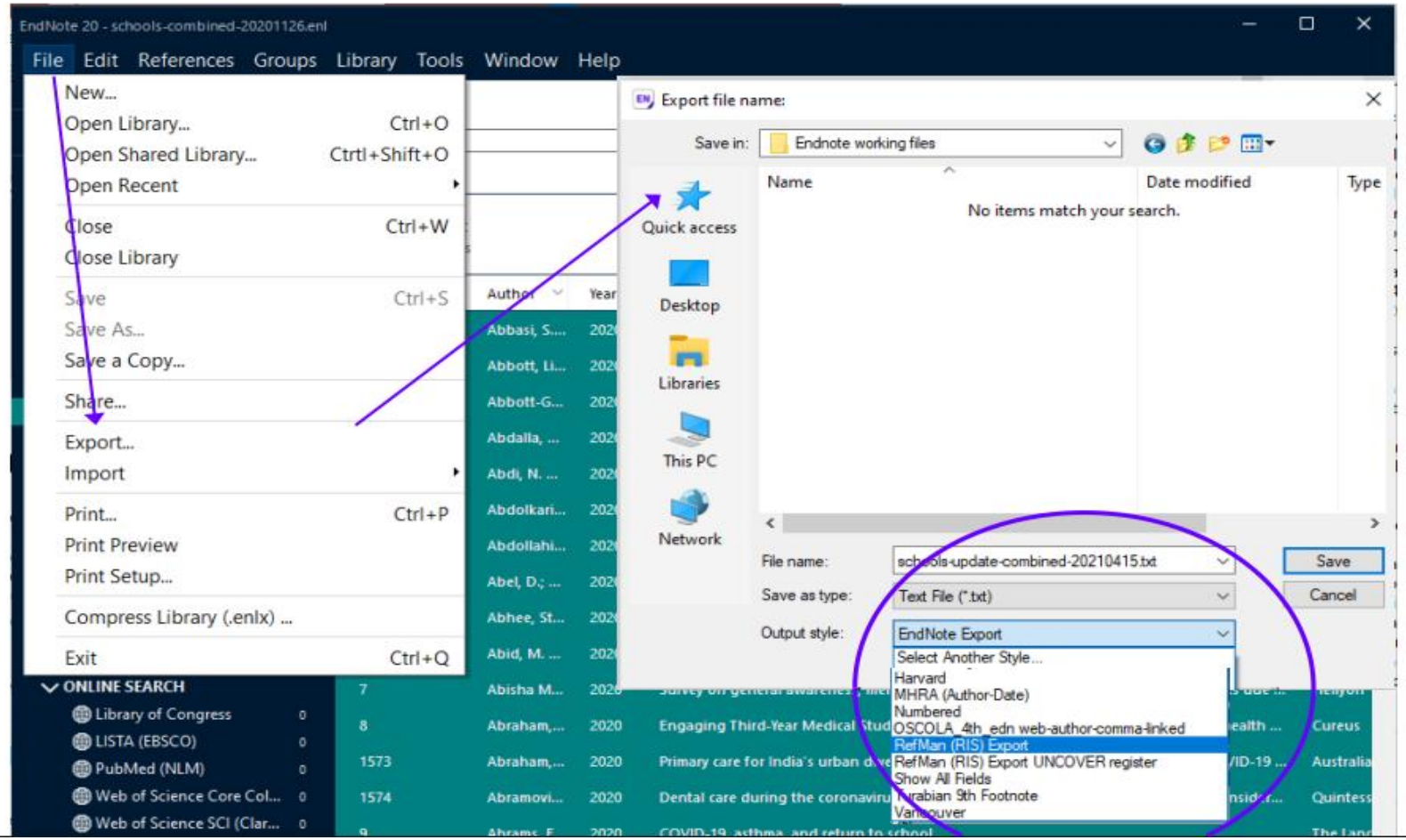


Screening 1: within EndNote



Screening 2: exporting in RIS or xml

Reference Manager (RIS) exported as text file type .txt is a standard format, but some tools (e.g. Covidence) may also work with EndNote export as file type .xml



Screening 3: Exporting to excel

For export to Excel, use the style, BasicExportToExcel

- Select all relevant records
- Right-click and Copy Formatted
- Paste into Excel using right-click menu
- Internal line breaks within the record text cause problems



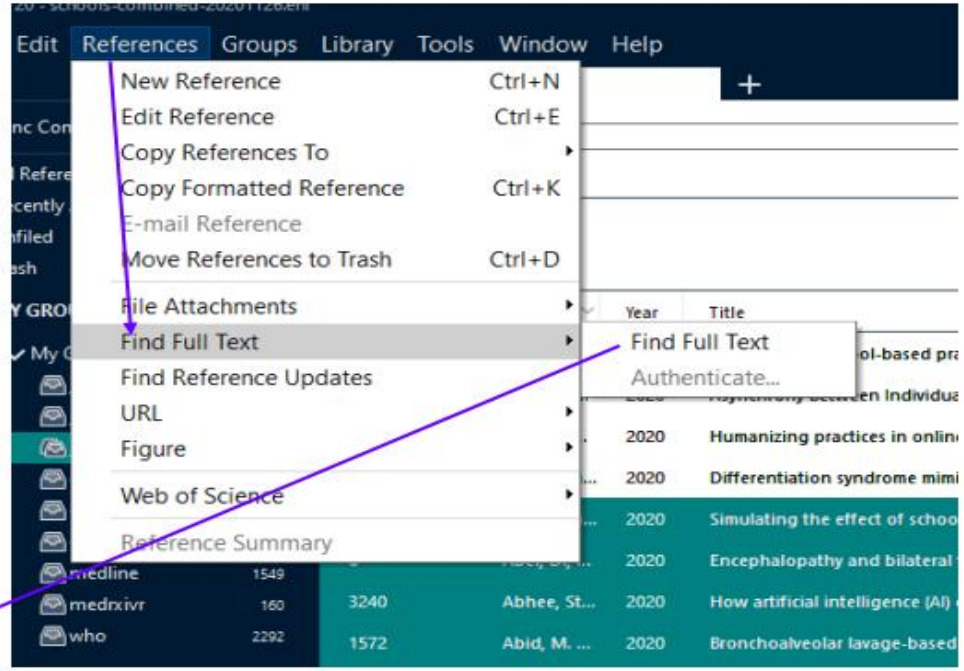
D	E	F	G	H	I	P	
Refere	Author	Year	e pub date	Title	Secondary Title	DOI	Abstract
Journal Ar	M. C. Alza	2020	19/04/2020	Severe COVID-19 during Pregnancy and Possible Vertical Transmission	Am J Perinatol	10.1055/s-	There are f (COVID-19) There is lin

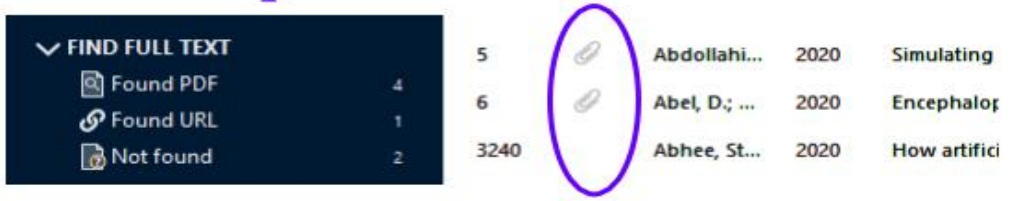
Find full text

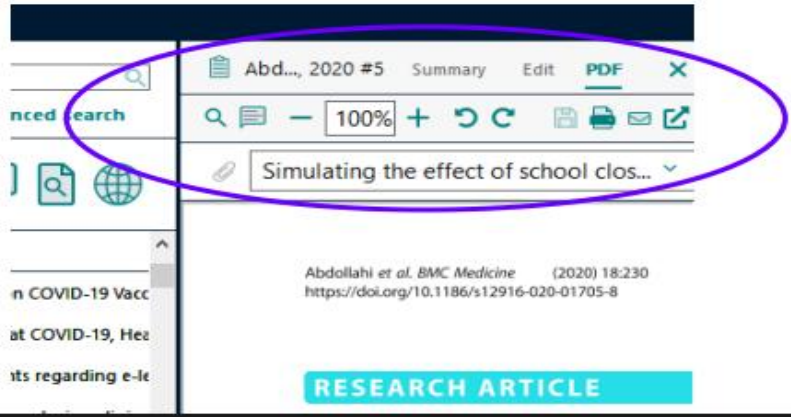
Save time!

Use UoE VPN for best results

Works best for journal articles with DOIs

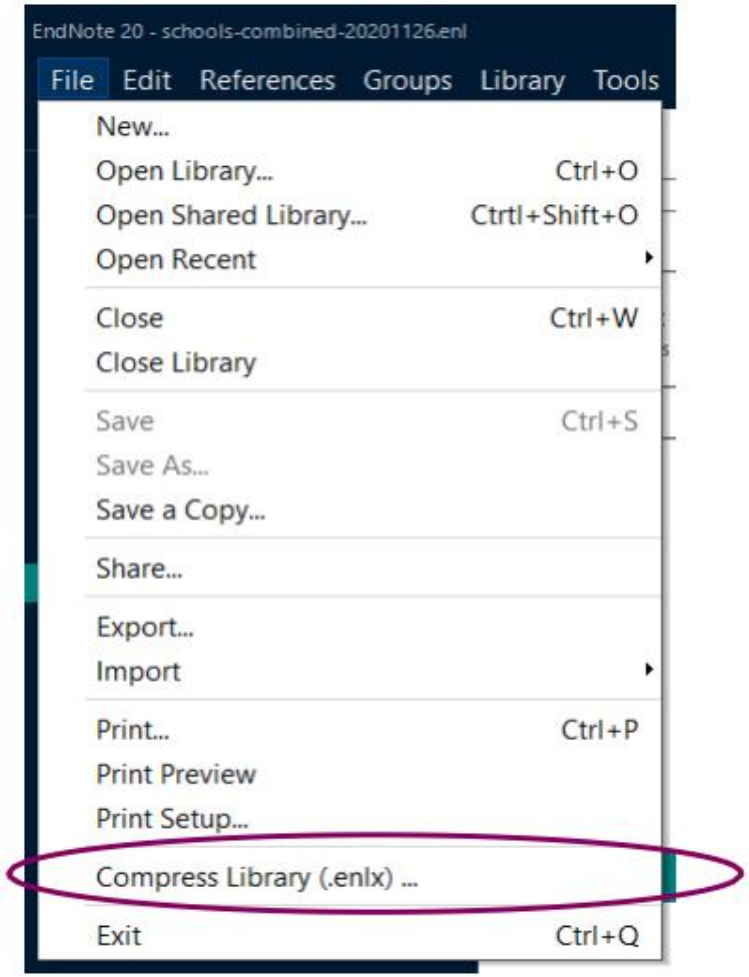






Backups

Create a compressed library at key stages and save somewhere that is backed up – e.g. OneDrive



Sharing with collaborators



Beware of using cloud servers for storing Libraries

Sharing goes via EndNote Online

Can share

- Use Groups in EndNote Online
- whole Libraries https://support.clarivate.com/Endnote/s/article/EndNote-Library-sharing-and-editing?language=en_US

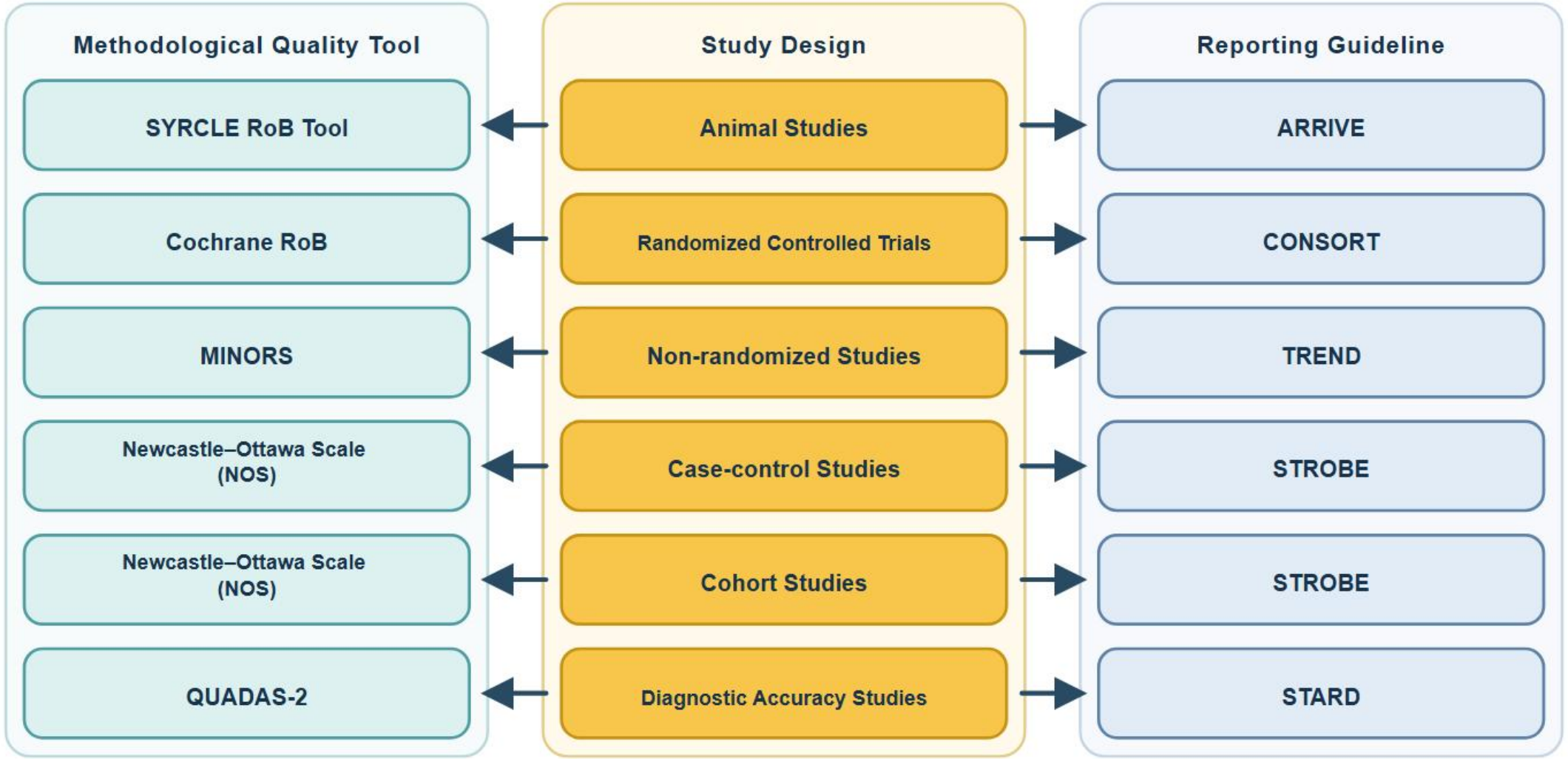
Alternative: Compressed Library on shared fileserver – nominate one team member to keep file version up to date

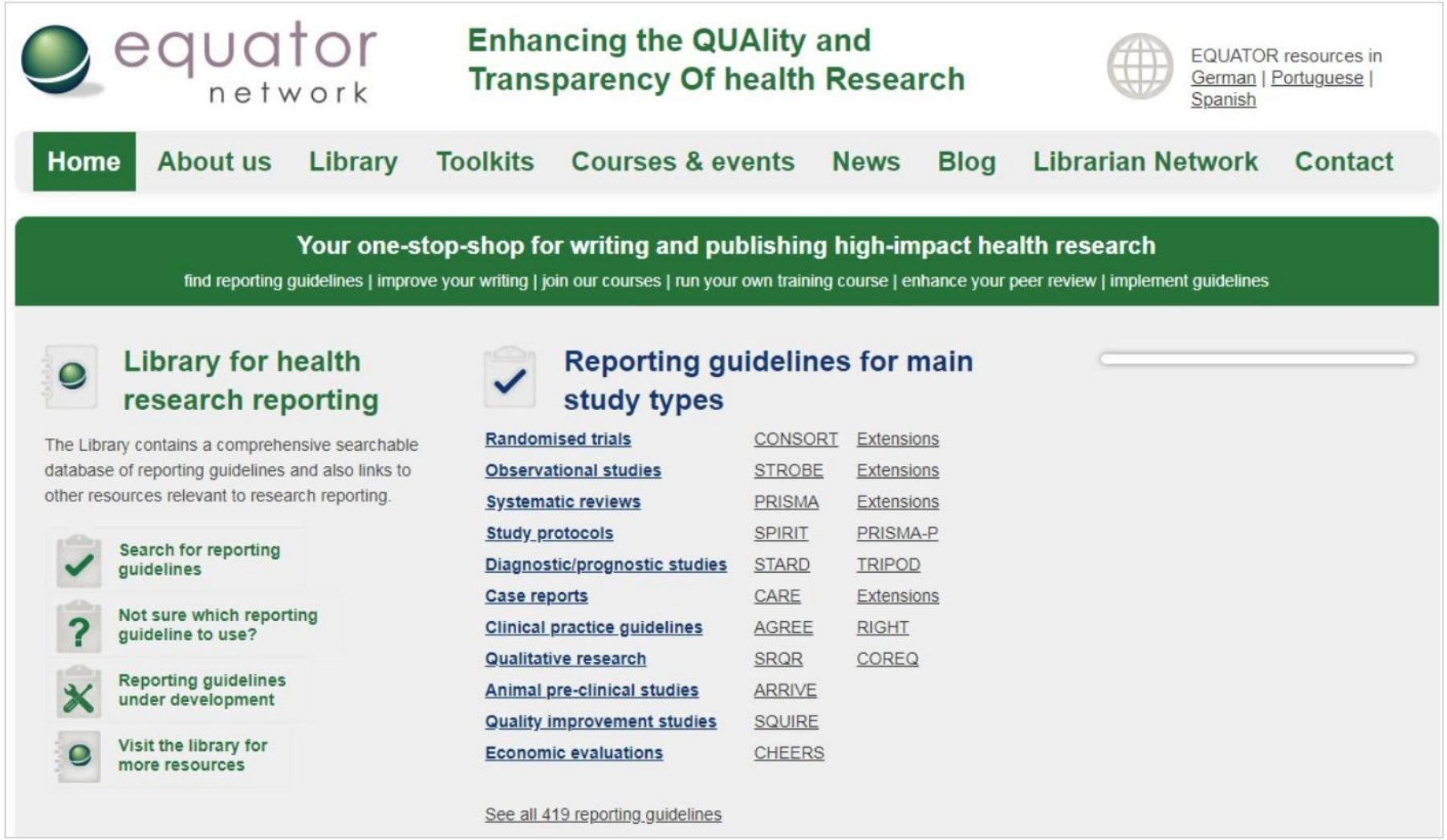
Key characteristics of a systematic review

- A clearly stated set of objectives with an explicit, reproducible methodology
- A systematic search that attempts to identify all studies that would meet the eligibility criteria
- An assessment of the validity of the findings of the included studies, such as through the assessment of risk of bias
- Systematic presentation and synthesis of the characteristics and findings of the included studies

[Liberati et al, *BMJ* 2007]

Quality Appraisal and Reporting Guidelines by Study Design




equator network
Enhancing the QUAlity and Transparency Of health Research
EQUATOR resources in German | Portuguese | Spanish
Home
About us
Library
Toolkits
Courses & events
News
Blog
Librarian Network
Contact
Your one-stop-shop for writing and publishing high-impact health research
find reporting guidelines | improve your writing | join our courses | run your own training course | enhance your peer review | implement guidelines
Library for health research reporting
The Library contains a comprehensive searchable database of reporting guidelines and also links to other resources relevant to research reporting.
Search for reporting guidelines
Not sure which reporting guideline to use?
Reporting guidelines under development
Visit the library for more resources
Reporting guidelines for main study types
Randomised trials CONSORT Extensions
Observational studies STROBE Extensions
Systematic reviews PRISMA Extensions
Study protocols SPIRIT PRISMA-P
Diagnostic/prognostic studies STARD TRIPOD
Case reports CARE Extensions
Clinical practice guidelines AGREE RIGHT
Qualitative research SRQR COREQ
Animal pre-clinical studies ARRIVE
Quality improvement studies SQUIRE
Economic evaluations CHEERS
See all 419 reporting guidelines

Taking account of study quality

- **Methodological quality of the primary studies may affect their results**
- **The influence of study quality should routinely be examined**
  - How?
- **Quality is difficult to define**
  - design, conduct and analysis of a trial?
  - clinical relevance?
  - quality of reporting?
  - all of these?

Focus on Randomised Controlled Trials (RCTs)

Quality scales and checklists

- **Many exist (>40)**
  - vary hugely in complexity and scope
- **Many combine trial quality with reporting quality and other aspects (e.g. sample size)**
- **Very popular: Jadad et al 1996**
  - Simple score from 0 to 5
  - Serious flaws: combines methodology and reporting
- **Comparison of high vs low quality trials depends**
  - Results depend strongly on which scale is used

[Jüni et al, *JAMA* 1999]

Rethinking the approach to assessing study quality

- **2005: Cochrane Collaboration initiative**
- **Rationale: quality scores are not helpful**
- **Focus should be on aspects with potential for bias, such as**
  - concealment of treatment allocation
  - blinding of outcome assessment
  - handling of patient attrition

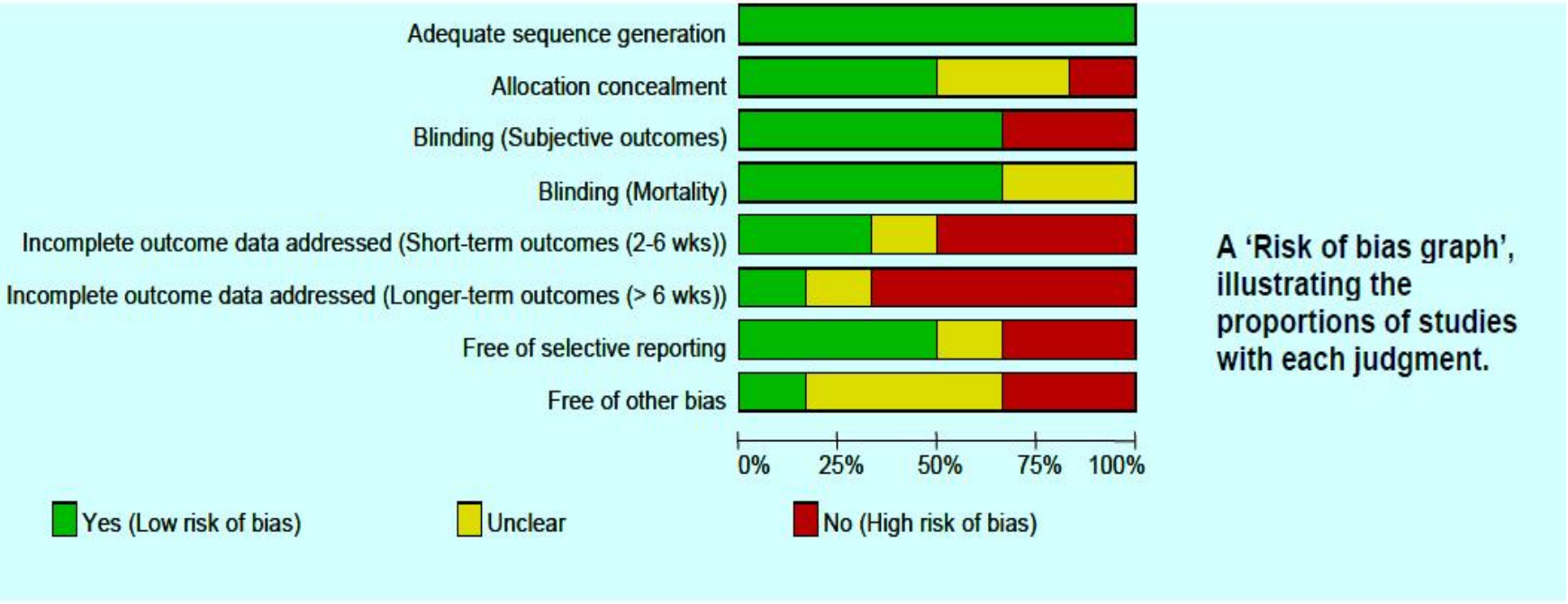
Cochrane Risk of Bias (RoB) tool

- The Risk of Bias (RoB) tool was developed over several years as a way to address concerns about the existing method of assessing study "quality" in Cochrane reviews
- Based on empirical evidence where possible
- It is intended to lead to uniform approach across reviews from all Cochrane review groups

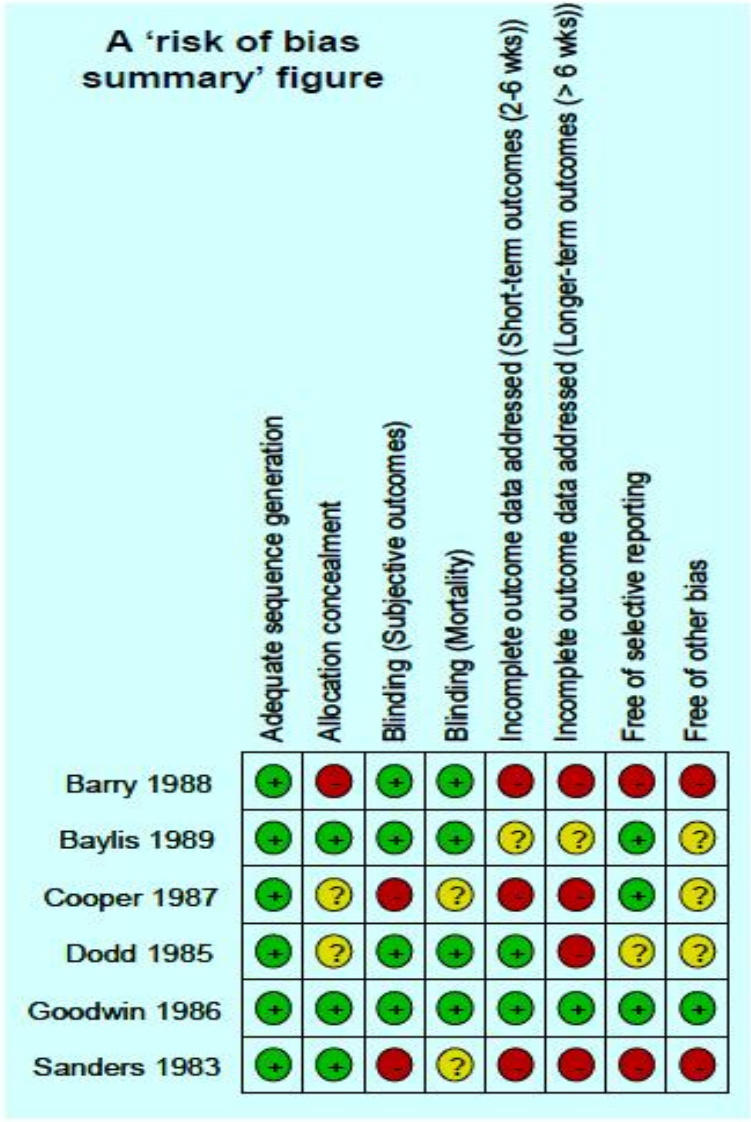
Developing a Risk of bias tool

- **In May 2005, 16 statisticians, epidemiologists, and review authors attended a three day meeting to develop the new tool**
- **Identified items that were truly potential biases rather than sources of heterogeneity or imprecision**
- **Seven areas were discussed leading to a proposed set of criteria for assessing protection from bias**
  - as adequate, inadequate, or unclear

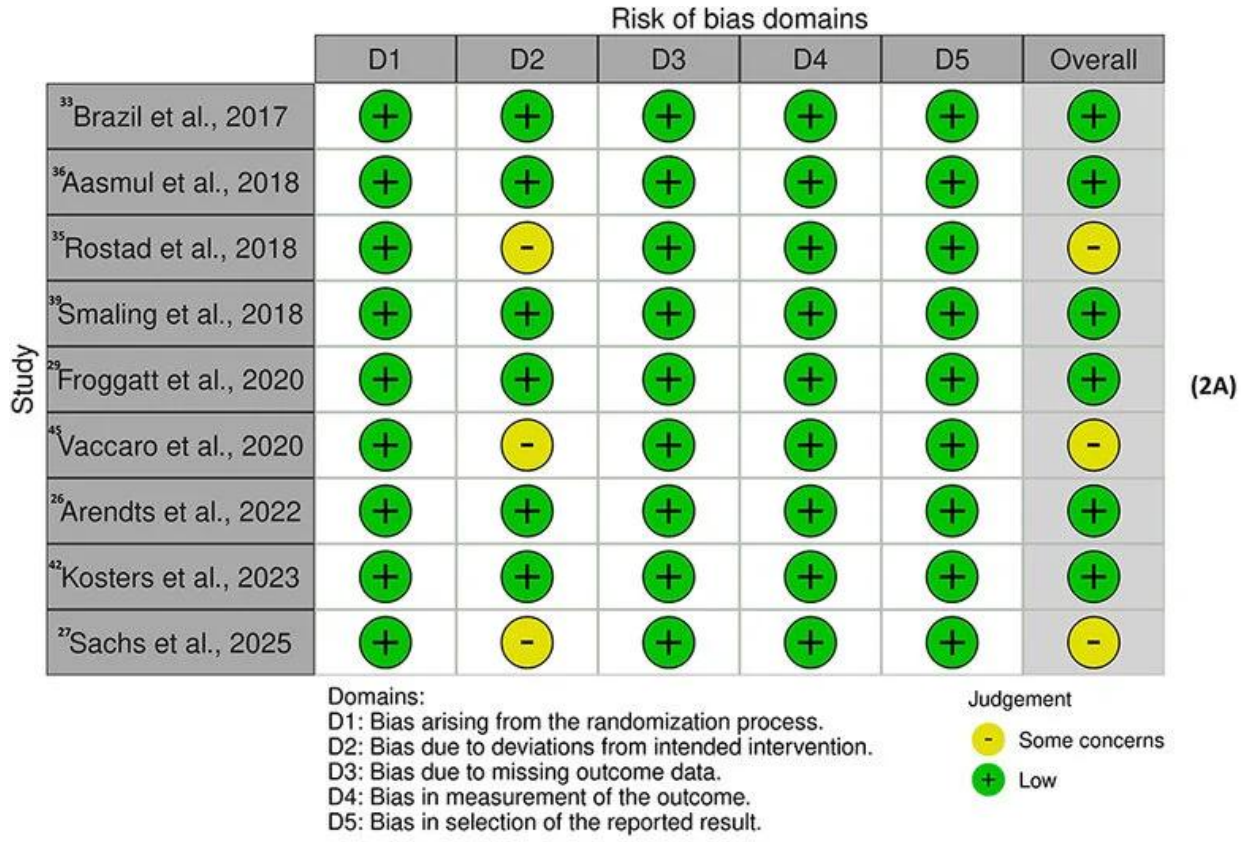
Cochrane Risk of Bias



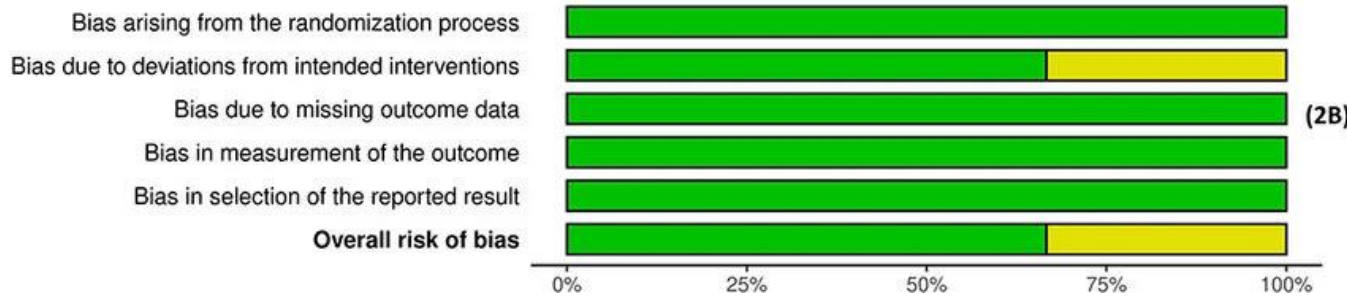
A 'Risk of bias graph', illustrating the proportions of studies with each judgment.



Cochrane Risk of Bias 2



(2A)



(2B)

What are we seeking to do in a RoB?



- Can I trust these results?

What are we seeking to do in a RoB?



- Is the methodology sound?
- Are the reported results therefore trustworthy?
- How trustworthy are the results?

Quality Assessment vs Risk-of-Bias Assessment

Related concepts with different purposes in evidence appraisal

Quality Assessment

SCOPE
Broad appraisal of study quality and reliability.

FOCUS
Study design, conduct, analysis, and reporting.

OUTCOME
Overall strength, relevance, and validity of the evidence.

VS

Risk-of-Bias Assessment

SCOPE
Focused appraisal of potential systematic error.

FOCUS
Bias domains that may distort study estimates or results.

OUTCOME
Trustworthiness and internal validity of the findings.

KEY DISTINCTION

Risk-of-bias assessment targets systematic error; broader quality assessment may include additional dimensions.

Difference Between ”quality assessment” and “Risk of Bias assessment”

Quality assessment → risk of bias assessment

- Bias = deviation from the truth
- E.g. an intervention has no effect, but the study shows an effect

Not precise, but accurate





Precise, but not accurate

Difference Between ”quality assessment” and “Risk of Bias assessment”

Quality Assessment

Common criteria assessed include:

- Appropriateness of Study design (e.g., randomized controlled trial, cohort study)
- Sample size
- Methods of data collection
- Statistical analysis
- Consistency of findings
- Generalizability of results
- Adequacy of follow-up
- Reporting clarity and transparency

Risk Of Bias

Key domains assessed for risk of bias typically include:

- **Selection bias**: How participants were selected and allocated to groups
- **Performance bias**: How interventions were implemented and whether there were deviations from the intended intervention
- **Detection bias**: How outcomes were measured and assessed
- **Attrition bias**: How incomplete data were handled and reported
- **Reporting bias**: Whether all expected outcomes were reported

Difference Between ”quality assessment” and “Risk of Bias assessment”

Quality Assessment

Outcome:

- Provides an overall judgement of how well the study was conducted.

Risk Of Bias

Outcome:

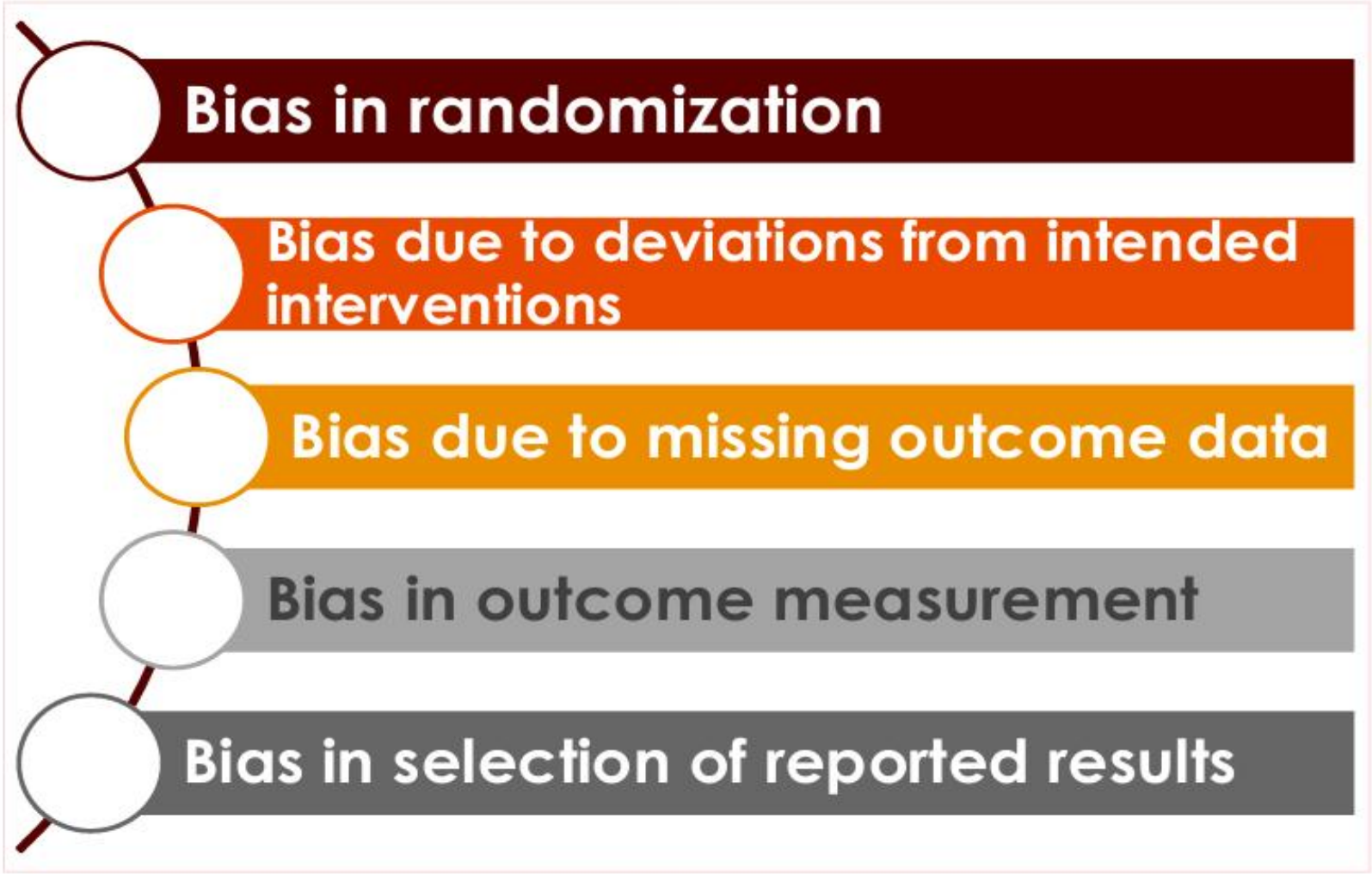
- Identifies specific areas where bias might have influenced the study results.

RoB 2 tool: **Principles**

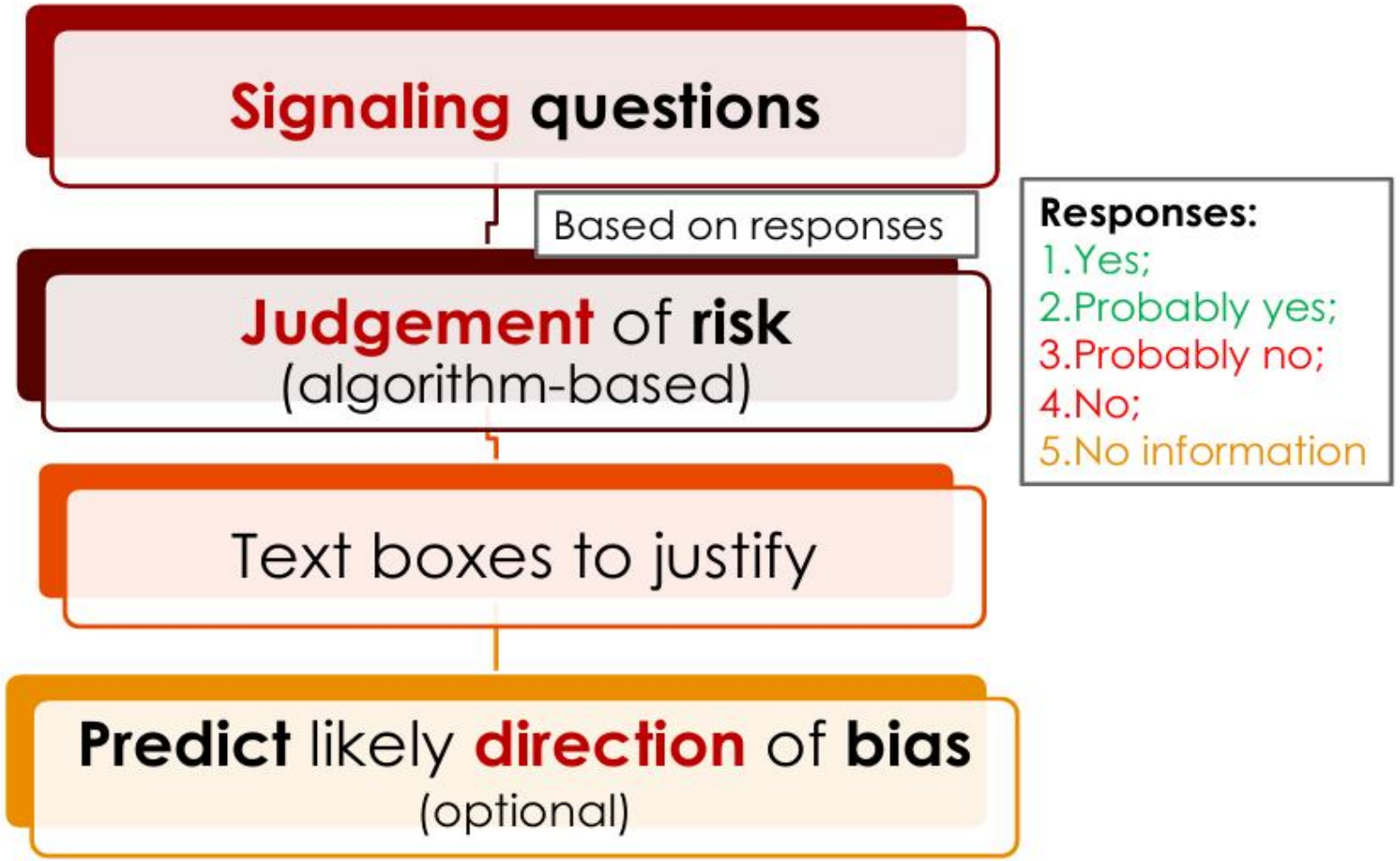
- Assess risk of bias (RoB) for **ALL** key **results** in **EACH study**
- Involves **judgement**; provide supporting data – should be **transparent**
- **Sources** of supporting **data** to be provided
- **Study-level** (e.g., bias in randomization)

Done by at least 2 people!

RoB 2 tool: **5 Domains**



RoB 2 tool: For each **Domain**



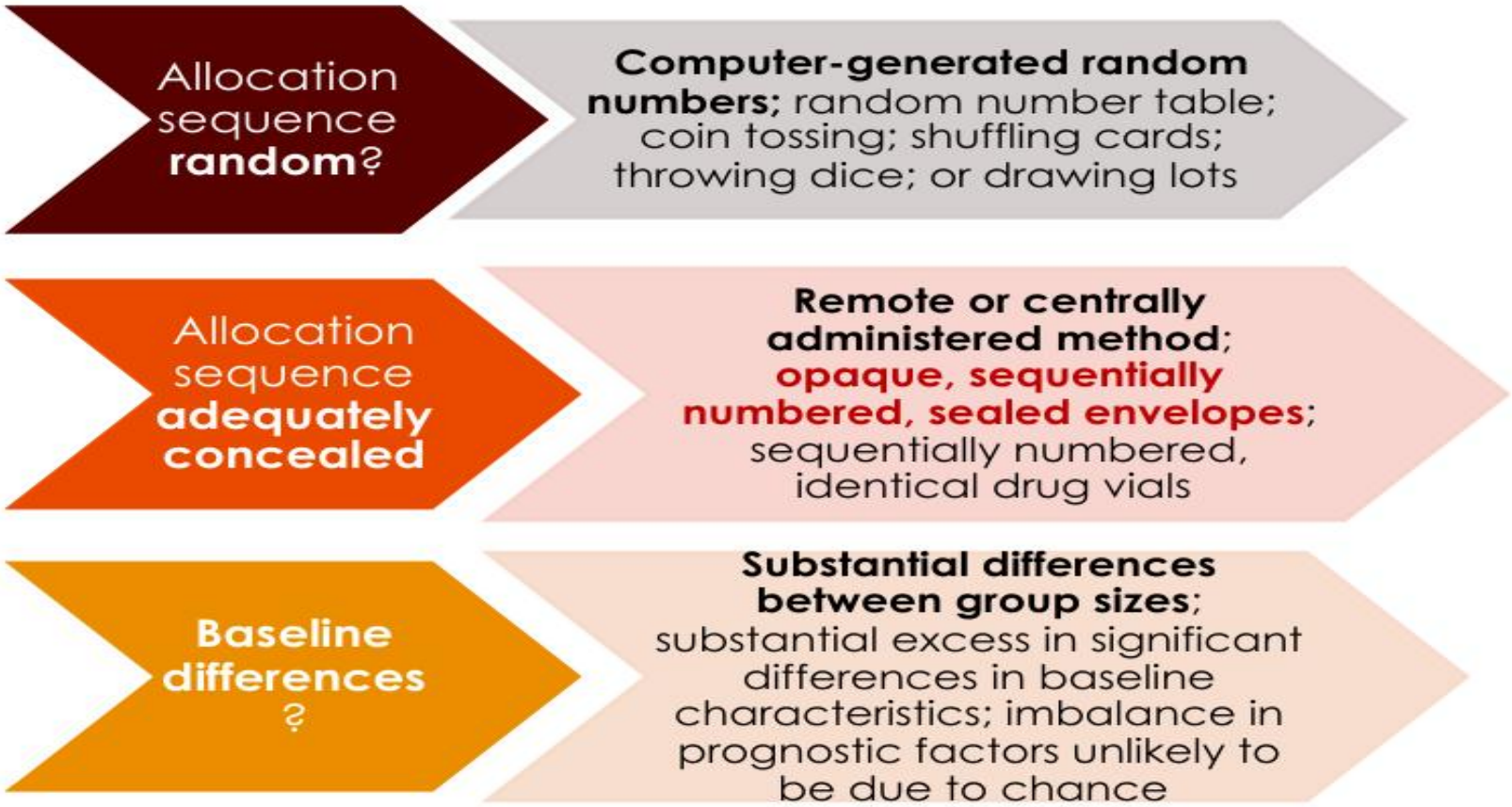
Domain 1: **Randomization**

Allocation sequence was **random**?

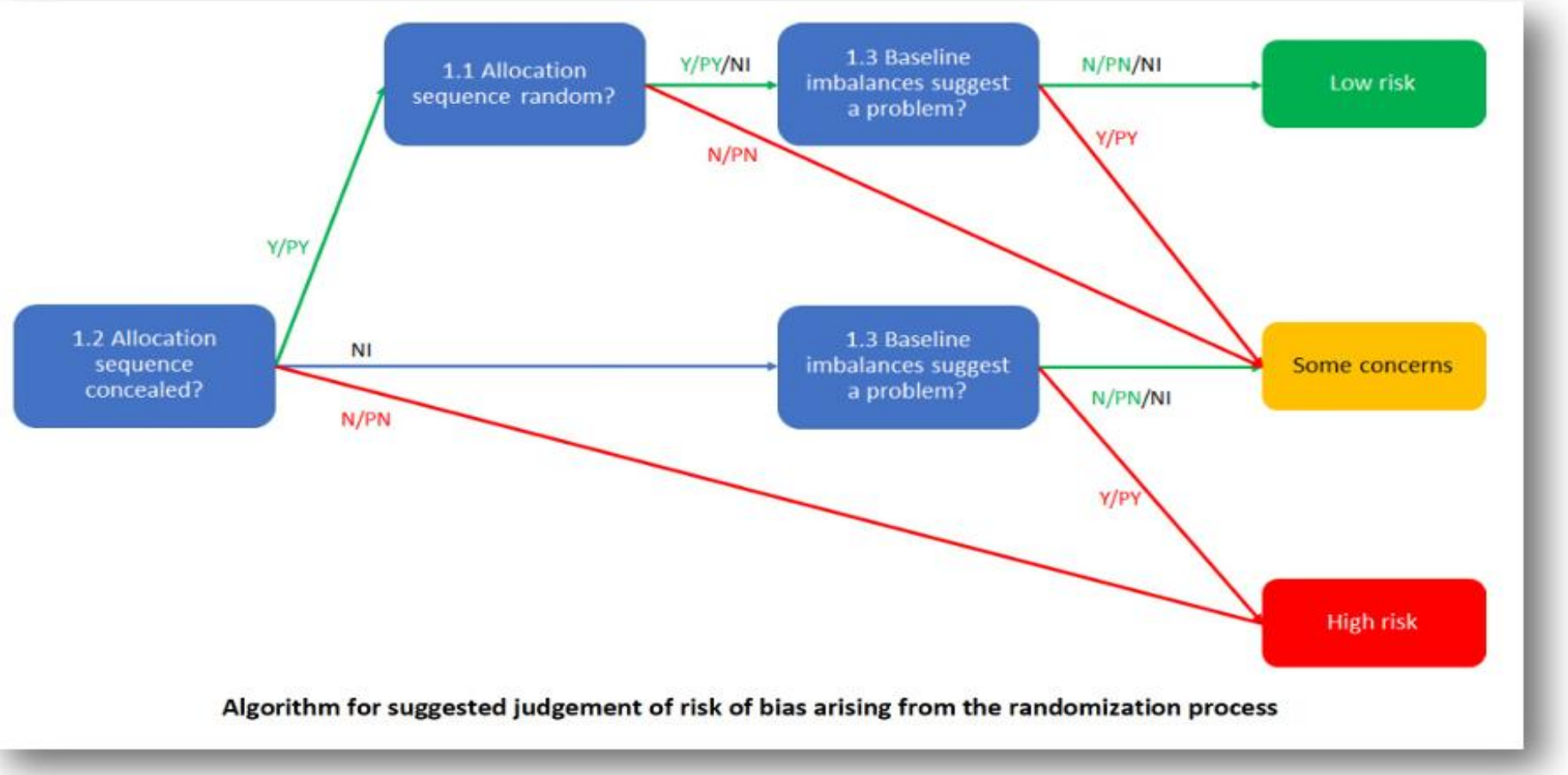
Allocation sequence was **adequately concealed**?

Baseline differences suggest **problem with randomization process**?

Domain 1: **Randomization**

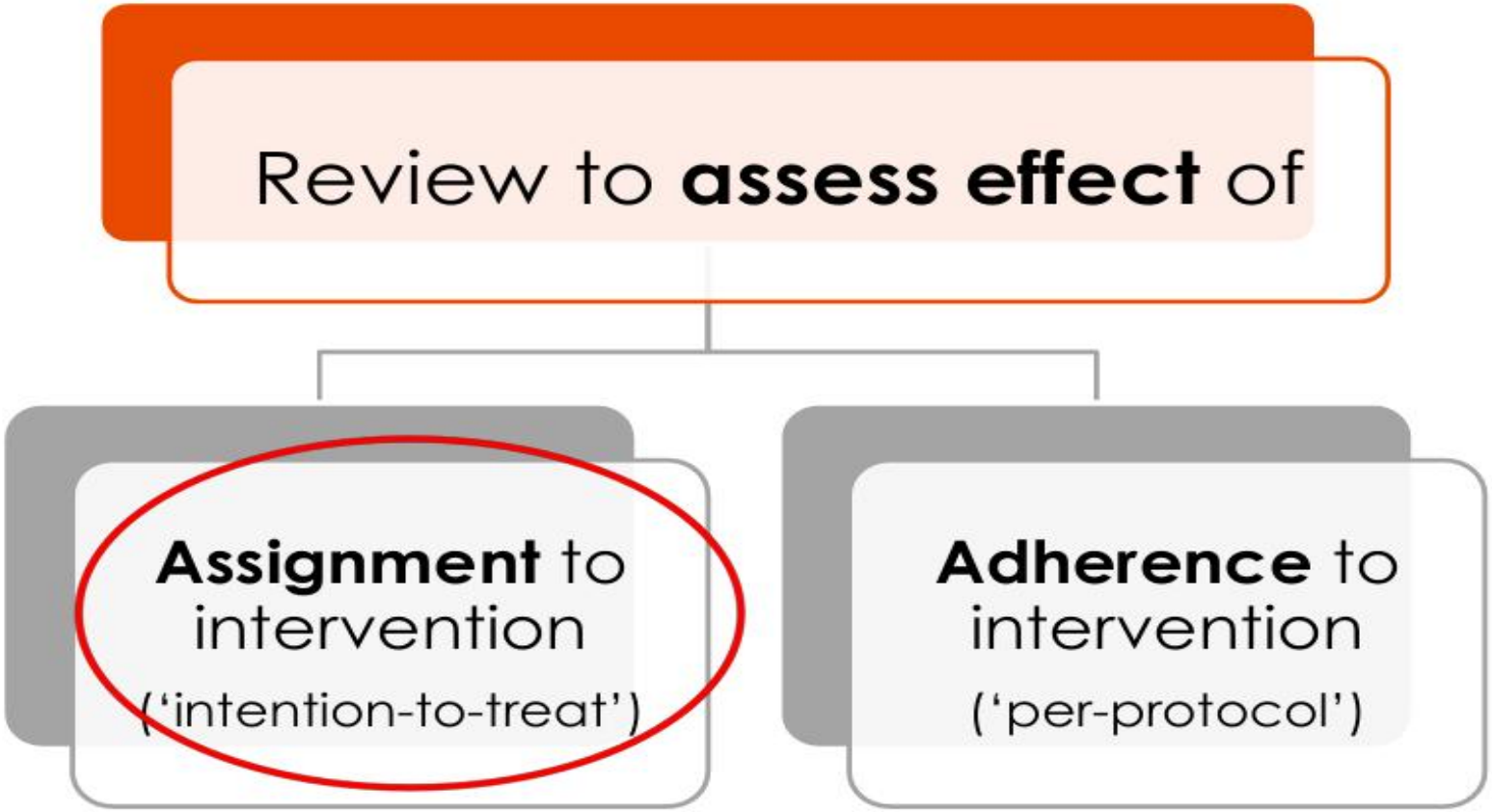


Domain 1: **Randomization**



Algorithm for suggested judgement of risk of bias arising from the randomization process

Domain 2: **Deviations** from interventions



Domain 2: **Deviations…**

Participants aware of **assigned intervention**?

Carers and **people delivering** interventions **aware** of **assignment**?

Deviations from intervention because of **trial**?

Deviations likely to have **affected outcome**?

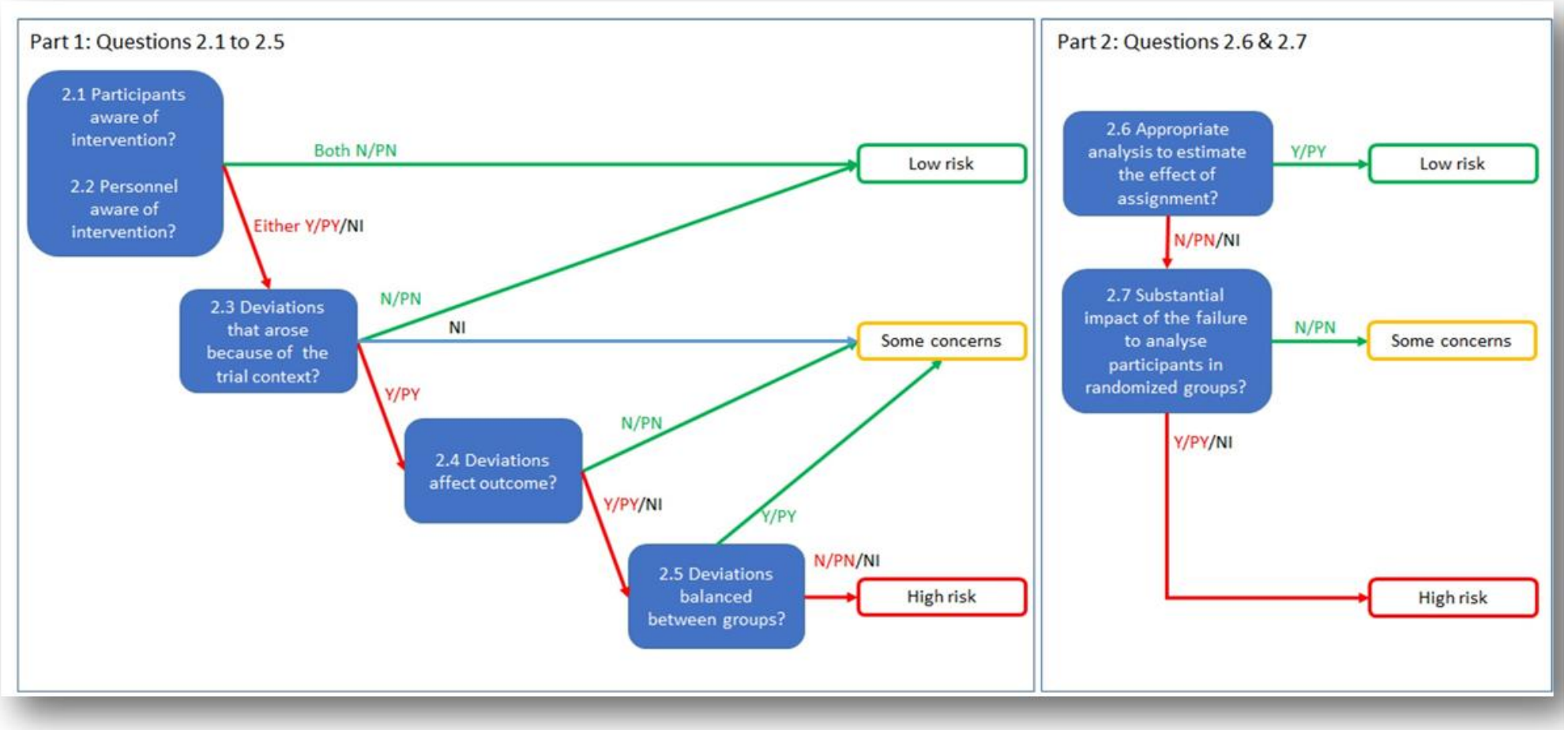
Domain 2: **Deviations…**

Deviations balanced between **groups**?

Appropriate analysis used to estimate **effect** of **assignment to intervention ?**

Potential for an **impact** of the **failure** to **analyse participants in the groups they were randomized**?

Domain 2: **Deviations**



Domain 3: **Missing outcome**

Data available for **ALL participants**?

Evidence that **result** was **NOT biased** by **missing outcome** data?

Could missing outcome depend on its true value?

LIKELY that missing outcome depended on its true value?

Domain 3: **Missing outcome**

Data available for **ALL**?

Number of participants with missing outcome is **sufficiently small** that their outcomes could have made **no important difference**

Result NOT biased by **missing outcome**?

Evidence may come from: (1) **analyses that correct for bias**; or (2) **sensitivity analyses** showing that results are little changed under a **range of plausible assumptions**

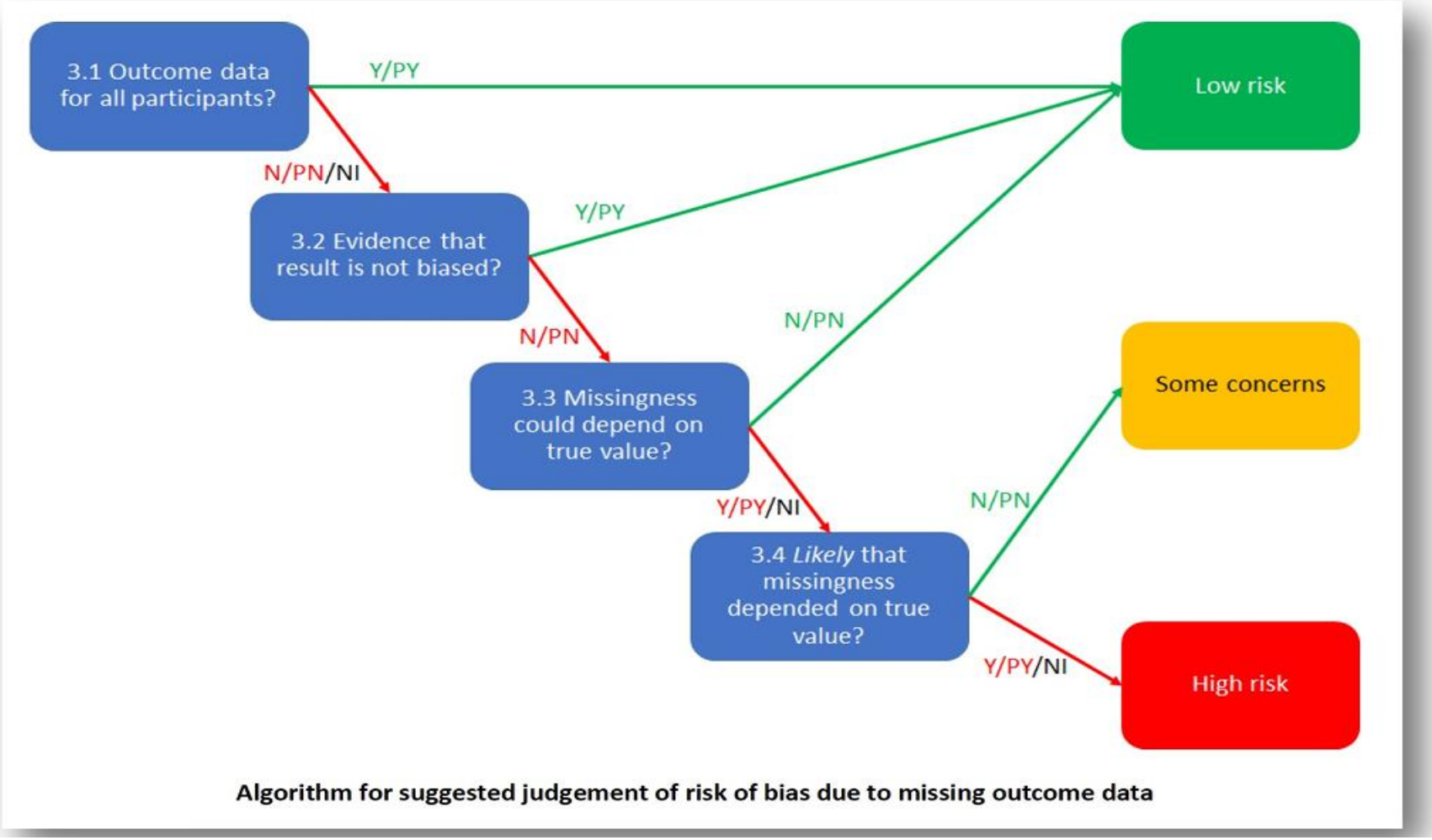
Could missingness depend on its **true value**?

If **loss to follow up**, or withdrawal from the study, could be **related to participants' health status**

Domain 3: **Missing outcome**



Domain 3: **Missing outcome**



Algorithm for suggested judgement of risk of bias due to missing outcome data

Domain 4: **Outcome measurement**

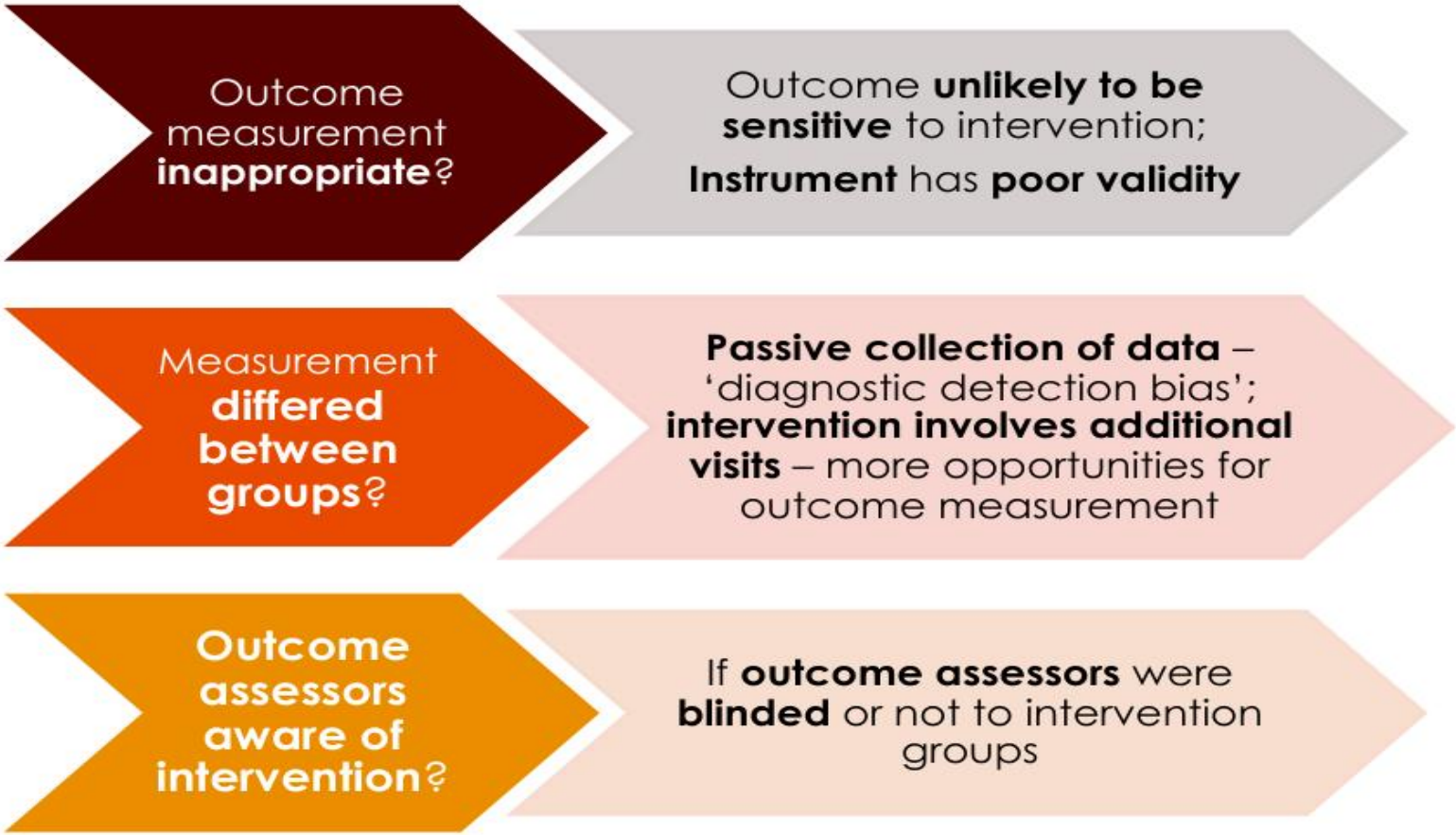
Method of outcome measurement **inappropriate**?

Measurement differed between the **groups**?

Outcome assessors aware of **intervention** groups?

Assessment influenced by **knowledge** of **intervention** received?

Domain 4: **Outcome measurement**

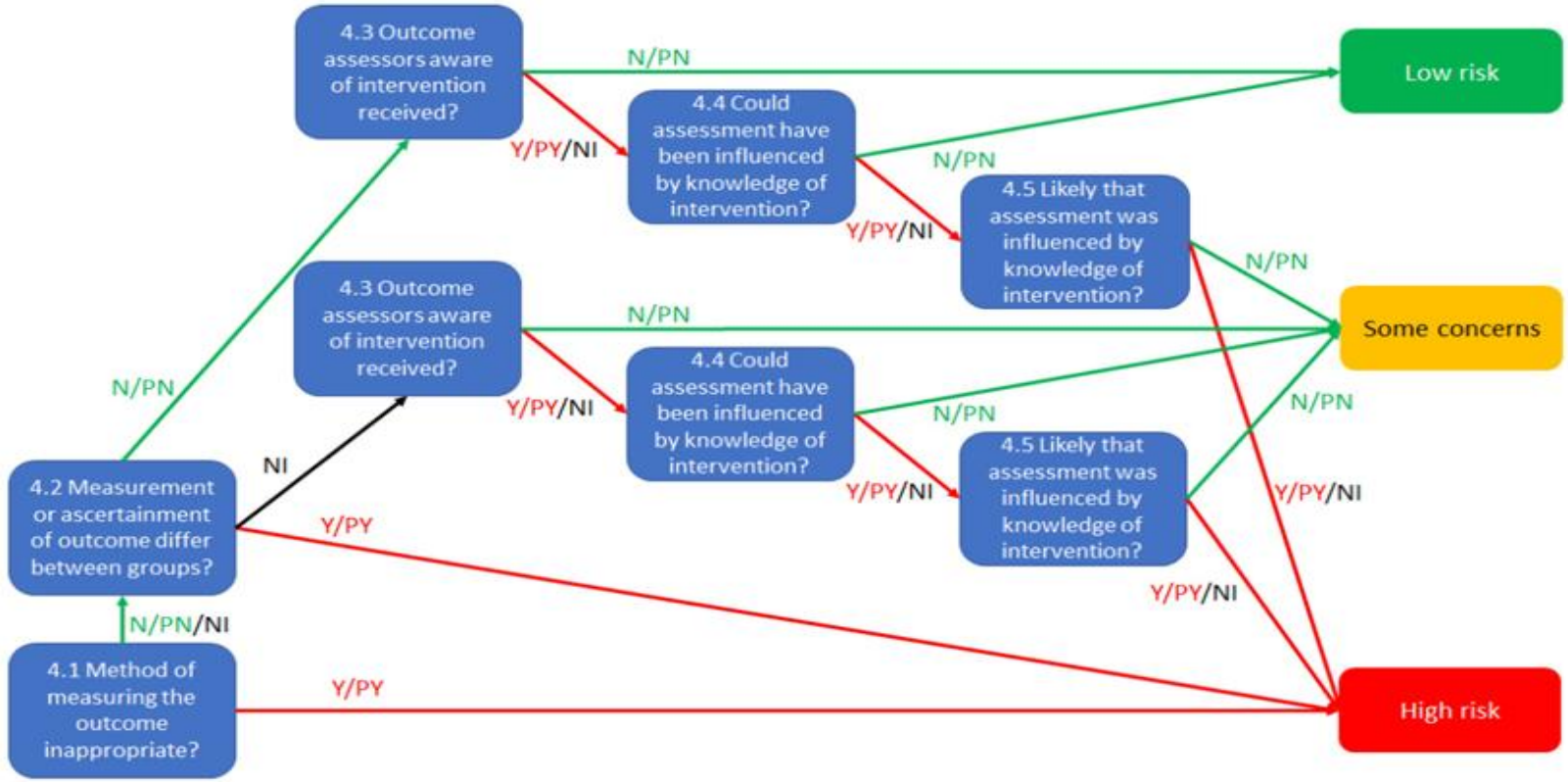


Domain 4: **Outcome measurement**

Outcome **assessment influenced** by **knowledge** of **intervention**?

Likely in 'subjective' outcomes and not in 'objective' outcomes like mortality

Domain 4: **Outcome measurement**



Algorithm for suggested judgement of risk of bias in measurement of the outcome

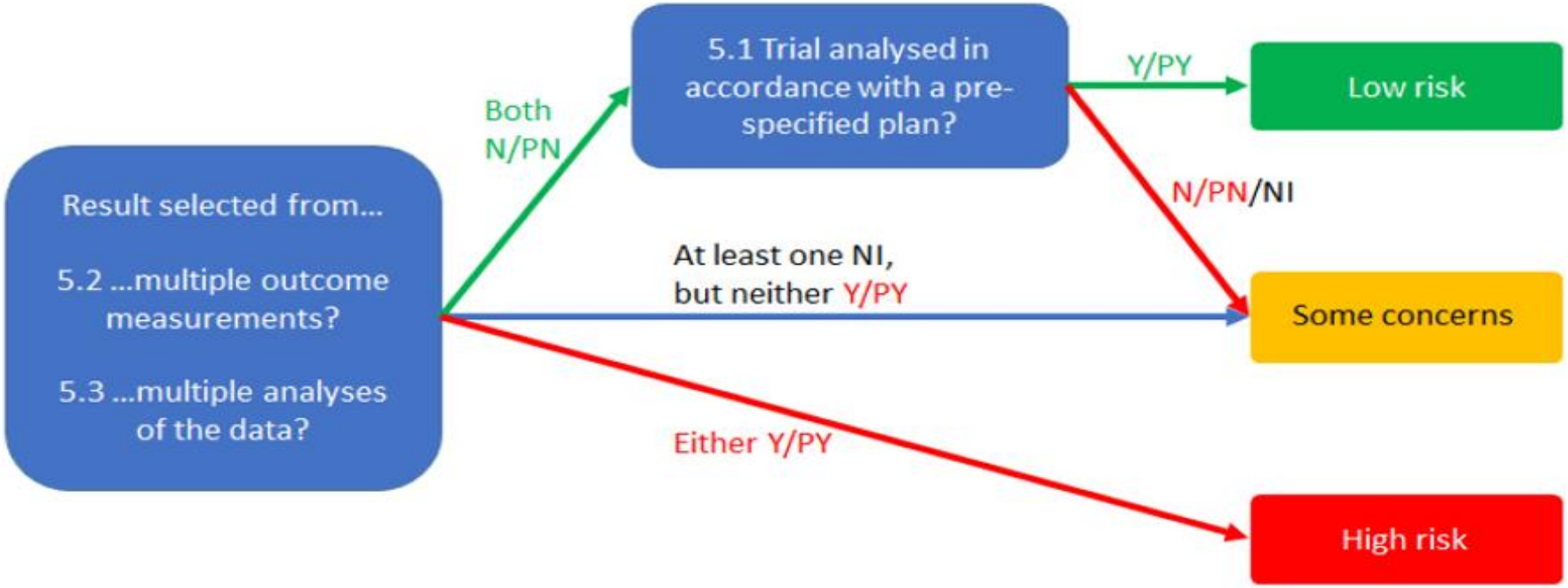
Domain 5: **Selection of reported result**

Data analysed as per the **pre-specified analysis plan**?

Result **selected from multiple outcome** measurements?

Result **selected from multiple analyses** of the **data**?

Domain 5: **Selection of reported result**



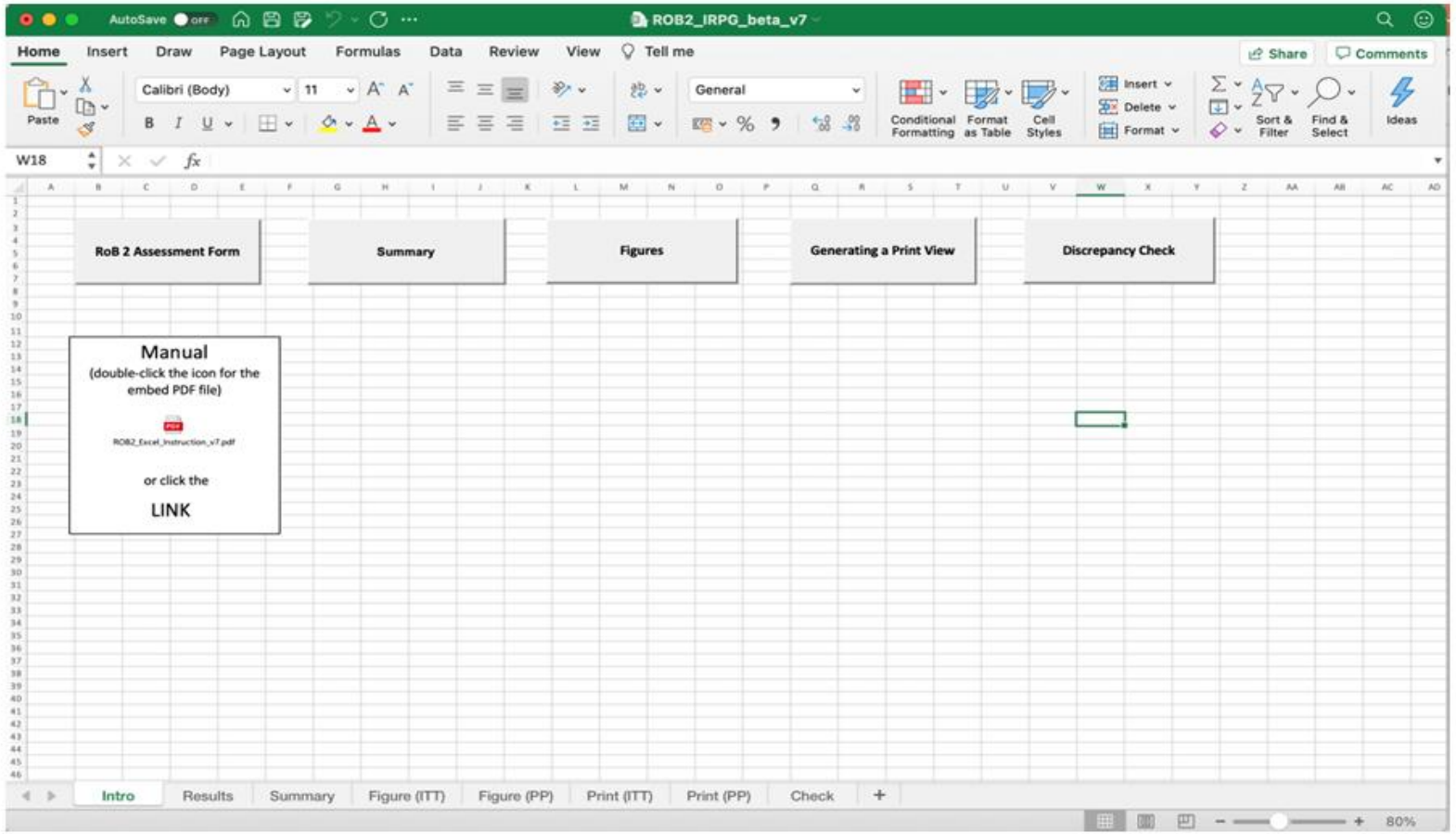
Algorithm for suggested judgement of risk of bias in selection of the reported result

OVERALL risk of bias

Table 1. Reaching an overall risk-of-bias judgement for a specific outcome.

Overall risk-of-bias judgement	Criteria
Low risk of bias	The study is judged to be at **low risk of bias for all domains** for this result.
Some concerns	The study is judged to raise **some concerns** in at least one domain for this result, but not to be at high risk of bias for any domain.
High risk of bias	The study is judged to be at **high risk of bias** in at least one domain for this result. Or The study is judged to have **some concerns** for **multiple domains** in a way that substantially lowers confidence in the result.

Excel tool

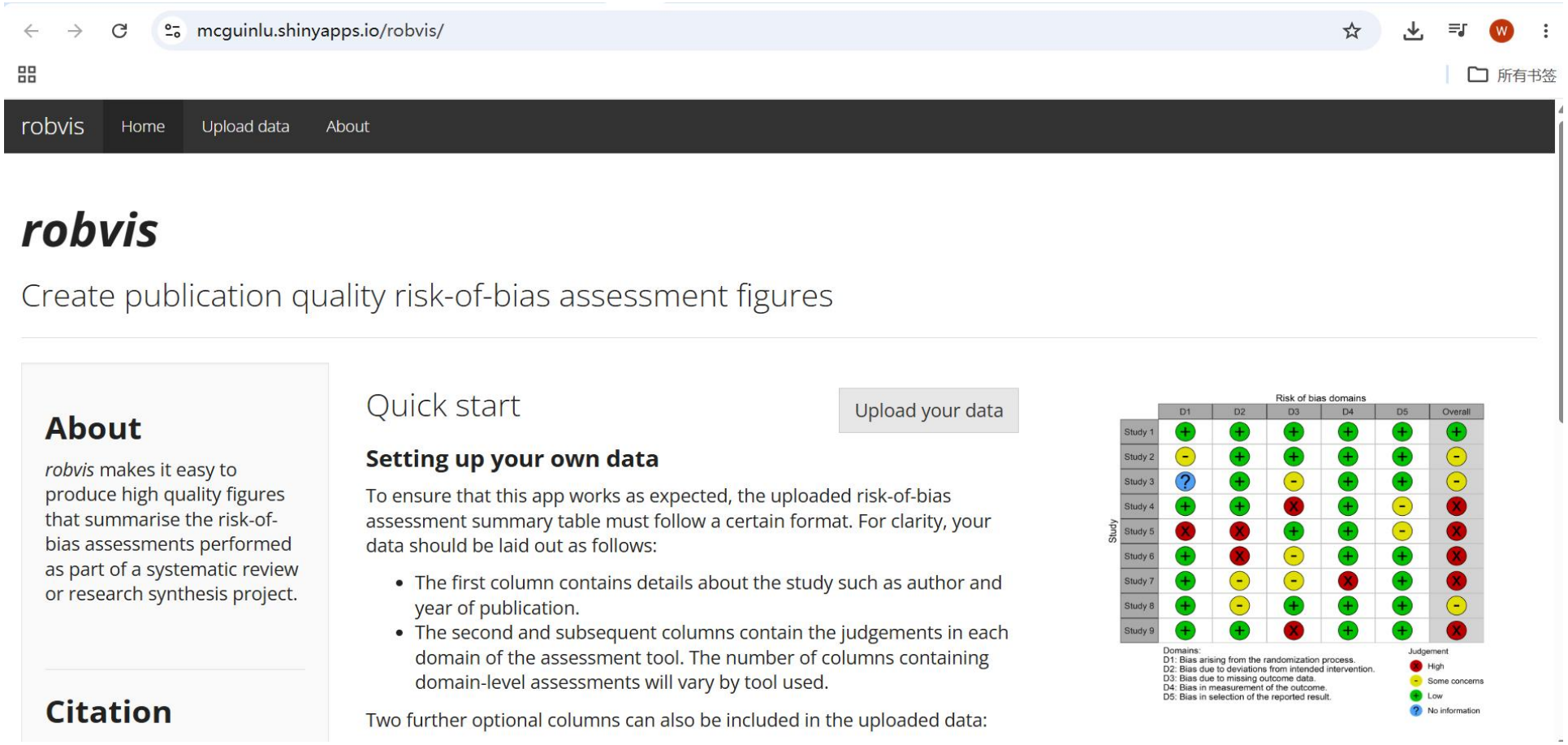


Cochrane

Simulated study description

- A total of 200 participants were randomly assigned in a 1:1 ratio to Drug A or a matching placebo, with 100 participants in each group. An independent statistician generated the allocation sequence using a computer. Allocation was implemented through a central randomization system, and recruiting personnel could not predict upcoming assignments. Baseline differences did not suggest a problem with randomization.
- Participants, personnel delivering the interventions, and outcome assessors were blinded to treatment allocation. Drug A and the placebo were identical in appearance, and blinding was maintained.
- All 200 participants completed the Week 8 blood pressure assessment and were analyzed in their originally assigned groups. No participants were excluded after randomization.
- Both groups were assessed using the same validated electronic blood pressure monitor and standardized measurement procedures.
- The protocol and statistical analysis plan were finalized before unblinding and specified the outcome, assessment time point, and analysis method. The reported result followed this plan, with no selection of measurements, time points, or analyses based on the findings.

robvis is a web app designed to for visualizing risk-of-bias assessments performed as part of a systematic review.



Cochrane

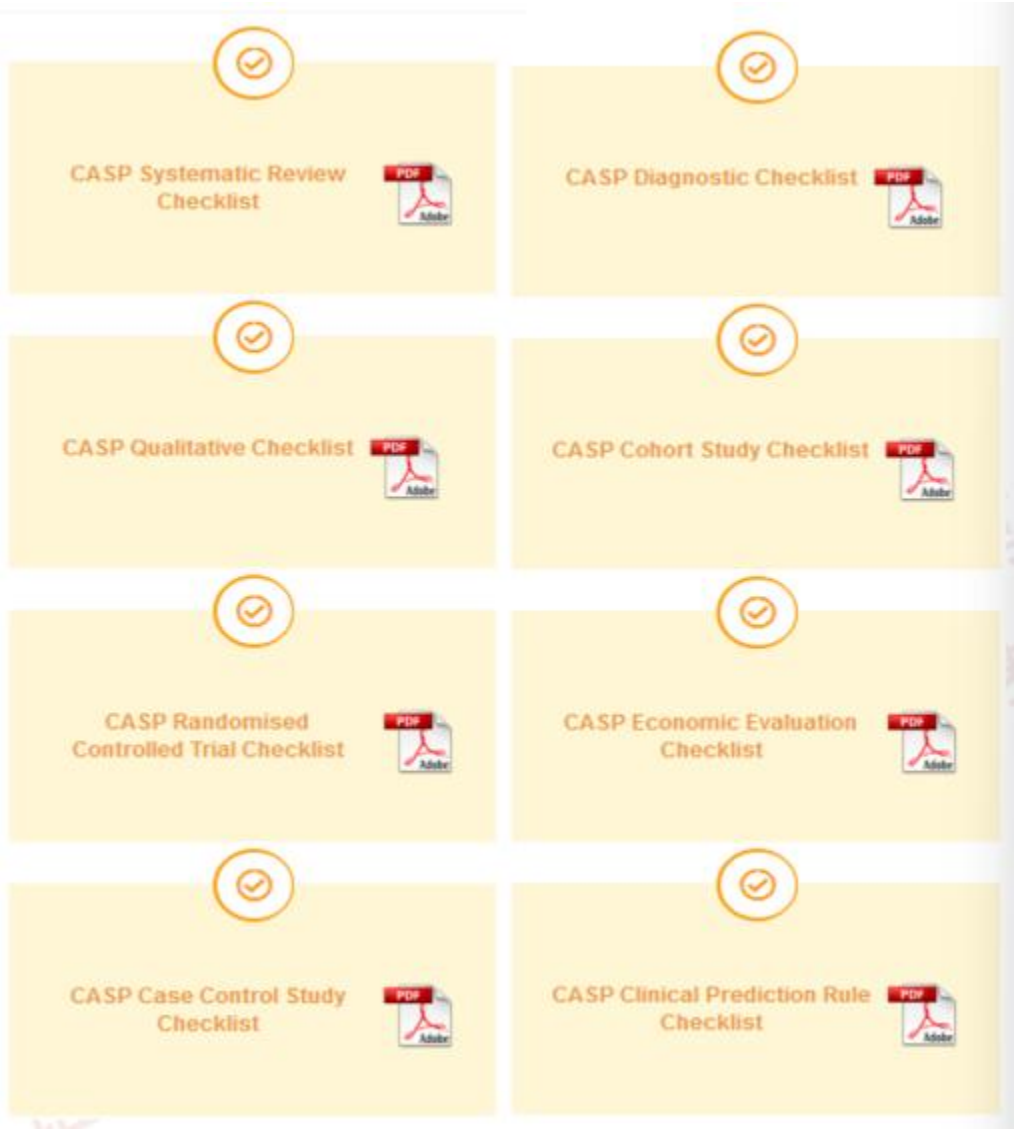
Modified Jadad Scale

Eight items	Answer	Score
Was the study described as randomized?	Yes	+1
	No	0
Was the method of randomization appropriate?	Yes	+1
	No	-1
	Not described	0
Was the study described as blinding? [a]	Yes	+1
	No	0
Was the method of blinding appropriate?	Yes	+1
	No	-1
	Not described	0
Was there a description of withdrawals and dropouts?	Yes	+1
	No	0
Was there a clear description of the inclusion/exclusion criteria?	Yes	+1
	No	0
Was the method used to assess adverse effects described?	Yes	+1
	No	0
Was the methods of statistical analysis described?	Yes	+1
	No	0

a: double-blind got 1 score, single-blind got 0.5 score.

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CASP Clinical Prediction Rule Checklist

Selective outcome reporting

- **Concern is that there is a tendency to selectively report outcomes that are statistically significant**
  - Empirical evidence [Kirkham *et al*, *BMJ* 2010]
- **Challenges in detecting such behaviour**
- **Ideally need study protocol**
- **May be internal evidence from publications or trials register**
- **Usually have to assess this item as "Unclear"**

Why bother about grading evidence?



- People draw conclusions about
  - Quality of evidence
  - Strength of recommendations

- Systematic, explicit approaches help
  - Facilitate critical appraisal
  - Protect against errors
  - Resolve disagreements
  - Communicate information

GRADE Working Group (2004)

Grading of **R**ecommendations **A**ssessment, **D**evelopment and **E**valuation

- System (and common language) that
  - was developed and updated by GRADE Working Group
  - has been endorsed by large number of organisations
  - expresses degree of confidence one can place in **quality of evidence** and **strength of recommendation**

- System for assessing quality of evidence of a body of evidence based on
  - study design
  - criteria for downgrading/upgrading

GRADE Working Group (2004);
Guyatt et al (2008)

Rating quality of evidence

GRADE quality of evidence

Definition: The degree of confidence in an estimate of effect.

High (++++)	We are very confident that the true effect lies close to that of the estimate of the effect.
Moderate (+++)	We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low (++)	Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect.
Very low (+)	We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of the effect.

Determinants of quality of evidence (1)

Study design:

- Bodies of RCTs start as high
- All other study designs start as low

Upgrading/downgrading criteria:

- Five factors can decrease quality of evidence
- Three factors can increase quality of evidence

RCTs versus observational study designs usually run through assessment separately

Determinants of quality of evidence (2)

Quality	Study design	Lower quality if:	Higher quality if:
High (++++)	Randomised controlled trials	Risk of bias (-1, -2) Inconsistency (-1, -2) Indirectness (-1, -2) Imprecision (-1, -2) Publication bias (-1, -2)	Large effect (+1, +2) Dose-response (+1) Direction of residual confounding and biases (+1)
Moderate (+++)			
Low (++)	Observational studies		
Very low (+)	Case reports, expert opinion, modelling		

Risk of bias (-/--)

- Risk of bias:
  Limitations in study design and execution with relevance to given outcome, based on quality appraisal of individual studies

- Indicators of risk of bias
  - Moderate or high risk of bias across most studies
  - etc.

- Lower quality of evidence
  - -1 if serious limitations in study design and execution
  - -2 if very serious limitations in study design and execution

Inconsistency (-/--)

- Consistency:
  Similarity of estimates of effect across studies

- Indicators of inconsistency
  - Differences in direction of effect
  - Variation in size of effect
  - Large I^2 value
  - etc.
- Distinguish between
  - **explained** heterogeneity (e.g. population, intervention, outcome)
  - **unexplained** heterogeneity

- Lower quality of evidence
  - -1 if large unexplained inconsistency
  - -2 if very large unexplained inconsistency

Indirectness (-/--)

- Directness:
  Extent to which populations, interventions, comparisons and outcomes are similar to those of interest

- Indicators of indirectness
  - Very different populations (e.g. age, sex, illness)
  - Surrogate outcomes
  - No direct comparisons
  - etc.

- Lower quality of evidence
  - -1 if serious uncertainty about directness
  - -2 if very serious uncertainty about directness

Imprecision (-/--)

- Precision:
  Is a consequence of sample size and number of events

- Indicators of imprecision
  - Small population (sparse data)
  - Small number of events
  - Wide confidence intervals around pooled effect (e.g. including RR=1)
  - etc.

- Lower quality of evidence
  - -1 if imprecise or sparse data
  - -2 if very imprecise or sparse data

Publication bias (-/--)

- Publication bias:
  Systematic under- or overestimate of effect due to selective publication of studies

- Indicators of publication bias
  - Small studies
  - Industry-sponsored studies
  - Asymmetric funnel plot
  - etc.

- Lower quality of evidence
  - -1 if publication bias is strongly suspected
  - -2 if publication bias is very strongly suspected

Three factors for upgrading

- Large or very large effects are less likely to be spurious
  - +1 if relative risk reduction ≤ 0.5 or risk ratio ≥ 2
  - +2 if relative risk reduction ≤ 0.8 or risk ratio ≥ 5
- Evidence of dose-response gradient
  - +1 if dose-response gradient observed
- If effect observed: All plausible residual confounding and biases would have reduced effect
- If no effect observed: All plausible residual confounding and biases would have increased the effect
  - +1 if appropriate direction of residual confounding and biases

Grading strength of a recommendation

GRADE strength of recommendation

Definition: The degree of confidence that desirable effects of adherence to a recommendation outweigh undesirable effects.

- **Strong** recommendation:
  The panel is confident that the desirable effects of adherence to a recommendation outweigh the undesirable effects.

- **Weak/ conditional/ discretionary** recommendation:
  The panel concludes that the desirable effects of adherence to a recommendation probably outweigh the undesirable effects, but is not confident.

Low quality of evidence, strong recommendation?

- Clear distinction between
  - quality of evidence, driven by confidence in body of evidence
  - strength of recommendation, driven by confidence in body of evidence plus other considerations
- All combinations possible, e.g.:
  - High quality of evidence -> weak recommendation
  - Very low quality of evidence -> strong recommendation
- Paradigmatic situations for strong recommendations in the absence of high-quality evidence (Alexander et al. 2016)

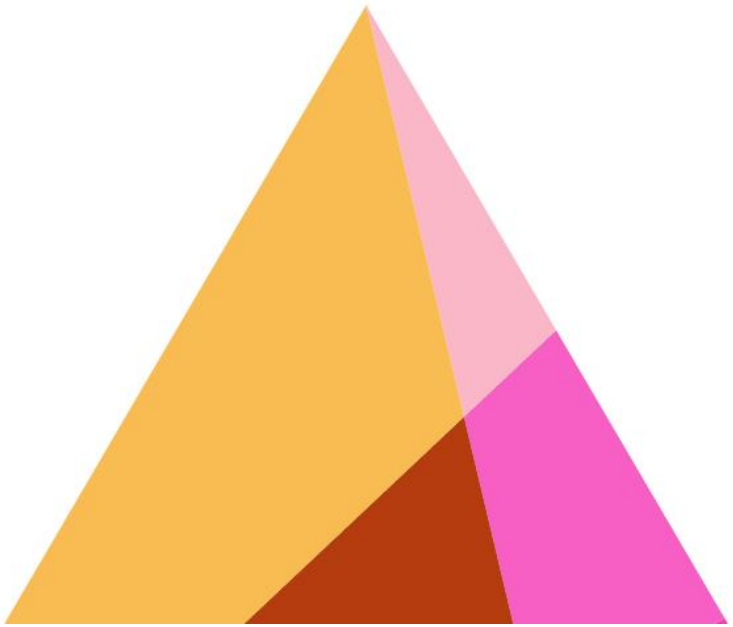
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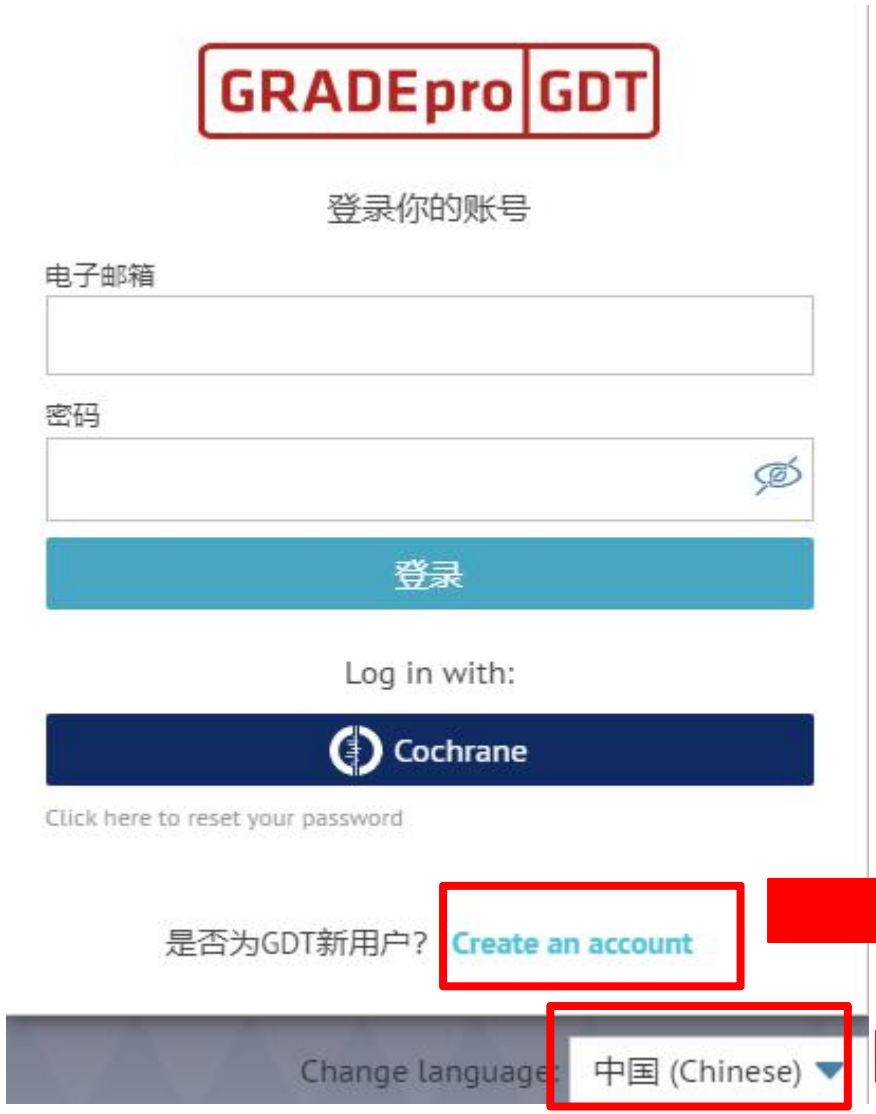
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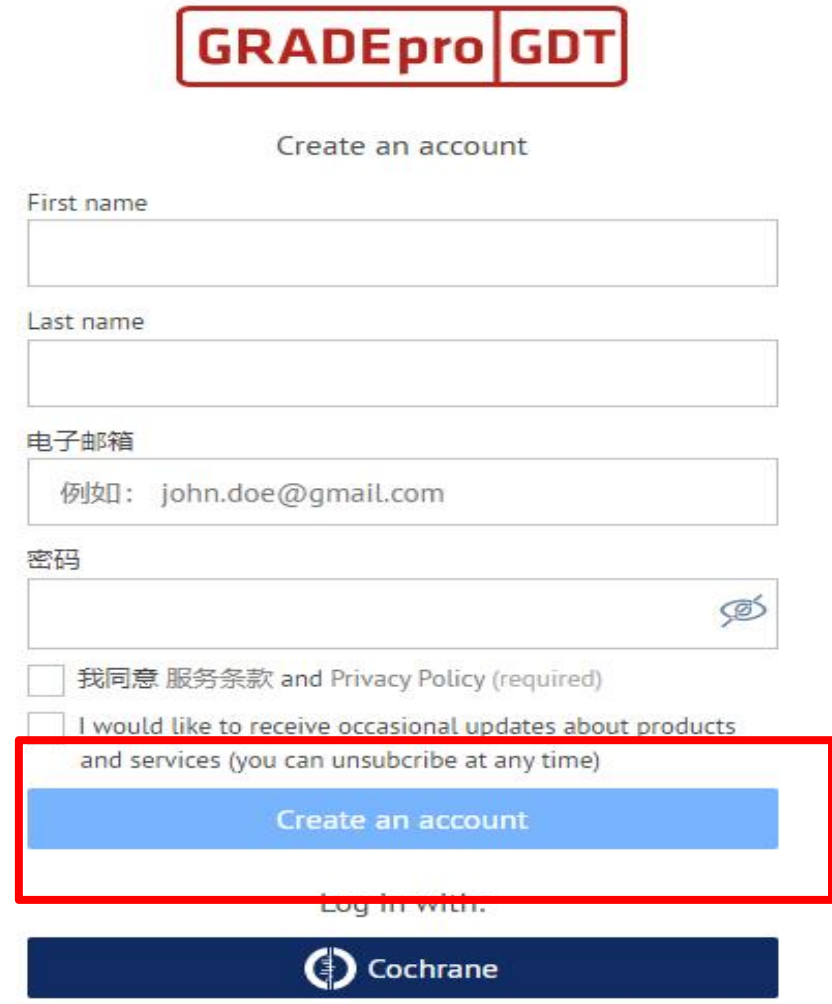
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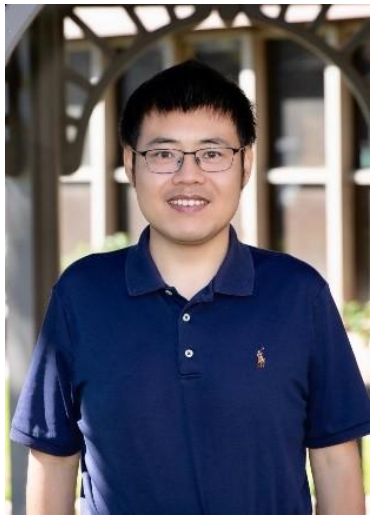

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